

Comparing Fixed and Adaptive Randomisation in Two-Arm Clinical Trials

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Abstract

The selection of a randomisation strategy in two-arm clinical trials with binary outcomes entails a fundamental trade-off between statistical efficiency and ethical responsibility. This report provides a systematic investigation of this trade-off through a Monte Carlo simulation framework implemented in R, where Monte Carlo methods serve as a computational approximation technique for estimating operating characteristics (including Type I error rate, statistical power, and ethical allocation) whose analytical forms are intractable under adaptive allocation mechanisms due to the dependence structure introduced by data-driven assignment.

Three individual-level designs are examined: Fixed 1:1 randomisation, Simple Response-Adaptive Randomisation (RAR), and Bayesian Response-Adaptive Randomisation employing Thompson Sampling within a Beta-Binomial conjugate model. The simulation framework is subsequently extended to cluster randomisation, where the relationship between the intracluster correlation coefficient and power loss is characterised via the design effect, and to Covariate-Adjusted RAR (CA-RAR), which incorporates logistic regression into the adaptive allocation mechanism to account for prognostic factors.

All types of designs have maintained the nominal Type I Error in null scenarios. The fixed 1:1 randomisation will be able to obtain the greatest amount of statistical power for a fixed total sample size as compared to the Bayesian RAR. In addition, the Bayesian RAR will allocate approximately 83% of the study population to the treatment with the largest effect size resulting in an approximate increase of 10 successful outcomes per 100 participants above the fixed allocations. The greater ethical advantages of the adaptive randomised design are associated with a decrease of 15 percent or less in the power of the design at $n = 100$; however, this disadvantage decreases significantly at $n = 500$ based upon sensitivity analysis.

Adaptive cluster designs clearly performed worse than their fixed counterparts when there were fewer clusters and CA-RAR provided ethical rather than inferential advantages over standard Bayesian RAR. Overall, it appears that no single design can dominate all operating criteria simultaneously and therefore the optimal design choice depends upon both the anticipated magnitude of treatment difference and the weighted priorities given to participant welfare vs. inferential precision.

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1 Introduction

Randomisation plays a major role in contemporary clinical trials as this method of randomisation will allow for unbiased comparisons of different treatments while also allowing researchers to control for potential known and unknown confounders of treatment assignment in all arms of the study [3, 6]. By controlling how treatment is assigned (i.e., by assuring that there is no systematic relationship between participant characteristics and treatment), randomisation provides a more reliable basis for attributing differences in outcomes to the treatments being compared. However, the choice of a particular randomisation strategy presents a key trade-off with respect to the two goals of clinical research: statistical efficiency and ethical responsibility. Fixed allocation designs are simple to implement and provide good balance in terms of the number of participants allocated to each arm at baseline and facilitate straightforward inference about the effects of the experimental treatments [7]. However, they can also continue to assign participants to the inferior treatment as evidence mounts showing that one treatment has greater efficacy than the other. As such, the use of fixed allocation designs raises a very significant ethical issue regarding the well-being of participants throughout the duration of the trial.

Adaptive randomisation strategies were developed to be a method of systematically reducing this tension [4, 5, 8]. The method of Response-Adaptive Randomisation is an example of one such strategy; the method adjusts the probability of allocation based on the responses to each intervention and as such increases the likelihood that subsequent allocations will be to the intervention which appears to be the most effective at a given point in time. As such RAR has two primary goals: to improve ethical allocation while still providing sufficient randomness so that valid statistical inferences can be made from the data collected. Bayesian Response-Adaptive methods are an extension of the Bayesian response adaptive approach; these Bayesian Response-Adaptive methods include an explicit model of the uncertainty in the effect of each intervention and the Bayesian Response-Adaptive methods use posterior distributions to update the probability of allocation based on the information gained from previous allocations. Therefore, Bayesian Response-Adaptive methods allow for sequential learning within the trial itself and link the statistical estimation with the decision making process.

Beyond individual allocation of treatment (beyond the individual), additional examples of cluster randomisation, and covariate adjusted adaptive randomisation exemplify the potential for various design decisions to affect both the inference based on trials, and the impact of those trials on participants [2, 10]. When treatments act at the group or population level (and when there is an expectation of participant contamination), cluster randomised designs may be appropriate. Covariate adjusted designs are used to create greater balance in terms of the distribution of prognostic factors among groups; they also may serve to reduce efficiency losses that could occur when using adaptive allocation schemes. These approaches demonstrate that randomisation is a design decision with both statistical implications, and ethical implications.

The purpose of this report is to analyse the statistical tradeoffs, as well as ethical tradeoffs associated with using either a fixed or adaptive form of randomisation in two arm clinical trials that have binary outcomes [1, 5, 9]. A Monte Carlo simulation method developed in R has been used to evaluate trial designs based on both null and alternative (i.e., non-null) treatment effects for each design. The operating characteristics of the trial designs, including the probability of type one errors, the statistical power of the tests, and the percentage of patients receiving the better treatment, were estimated as part of the evaluation process to assess the extent to which different forms of allocation strategies can be used to create a balance between ensuring reliable inference, while also protecting participant welfare. Through the combination of theoretical analysis and simulation-based results, the research aims to provide clarity regarding the specific circumstances under which use of an adaptive approach offers distinct advantages compared to use of a standard fixed allocation strategy; and, conversely, to identify those situations in which the benefits related to ethical considerations are achieved without compromising statistical validity.

2 Background and Literature Review

2.1 Clinical Trial Design and Randomisation

Randomised Controlled Trials (RCTs) are at the top of the hierarchy of clinical evidence [3, 6]. Randomisation provides a statistically independent relationship between treatment assignments and both observed and unobserved participant characteristics, so that systematic bias is eliminated in comparisons of treatment arms, and a rationale exists for making causal inferences without requiring parametric assumptions about the distribution of all possible outcomes [1, 3]. Randomising participants into a two-arm trial, using n participants and randomly allocating them to either receive Treatment A or Treatment B, allows inference to be made from the joint distribution of the outcomes and assignments generated by the randomisation process [7]. The framework of RCTs is particularly appealing because the randomisation distribution is entirely known by the designers of the study; thus, the validity of statistical testing and construction of confidence intervals is not contingent upon the correctness of any outcome model [1].

Fisher provided the intellectual foundations of randomisation as a tool for making scientific inferences, based on his clear position: under the null hypothesis of no treatment effect, the distribution of any test statistic can be determined from the randomisation mechanism alone, without reference to any model of the population from which the participants were drawn [1]. Thus, the randomisation-based justification for inference is separate from the population-based justification underlying classical frequentist statistics. In a randomisation-based context, the p -value estimates the probability that a test statistic as extreme as the one observed could occur purely by chance under the randomisation scheme, based solely on the observed data. This distinction is not merely philosophical; it has practical implications for the selection of test statistic and interpretation of results, especially in trials of small size where asymptotic approximations may not be reliable [7].

In addition to being an appealing statistical framework for assessing the relative effectiveness of treatments, the type of randomisation strategy employed is also a critical design decision that will influence the statistical properties of the trial and the welfare of the participants involved, in ways that are likely to be counterproductive to each other. The degree to which these conflicting objectives are realised is the subject of the current report, and represents an issue that has been widely discussed in the clinical trials literature [4, 8]. The conflict is most acute when the two treatments being compared have dramatically different expected effects: in such cases, the costs associated with fixed equal allocation will be greatest, because a high percentage of participants will inevitably be allocated to the less effective treatment even as evidence of its inferiority grows [4].

Regulatory agencies exemplify the dilemma. For confirmatory trials, fixed equal allocation is the default, and any adaptive deviation from it must be justified statistically and ethically [7, 8]. Even though the justification requirement is sensible, adaptive allocations do present analytical difficulties, raise concerns regarding potential operational bias if accumulating data are not adequately masked, and may lead to inflated Type I error rates if they are not managed carefully [8]. Adaptive allocations are very appealing on ethical grounds in those situations where interim data indicate that one treatment is superior. Just because it is not always clear which treatment is superior, does not mean that some other treatment cannot be considered inferior [4, 7].

The report compares the operating characteristics that apply to fixed and adaptive allocation, Type I error rate and power under simulation in order to investigate when one might be preferable to the other. The binary outcome setting was chosen for mathematical tractability since it can be modelled with a conjugate Bayesian Beta-Binomial model and also of practical interest since the most common primary endpoints in Phase II and Phase III trials are binary endpoints such as response, remission and survival at a fixed timepoint. By using binary endpoints, one can have a simulation study that is entirely self-contained and transparent: the data generating mechanism is completely specified by giving values for two parameters θ_A and θ_B ; the test statistic is the standard two-sample proportion test; operating characteristics can all be estimated to as high a level of precision as desired by increasing the number of Monte Carlo replicates [3, 7].

2.2 Fixed Allocation Methods

When using fixed allocation methods for randomising study subjects to experimental treatments, the method used to allocate participants to a treatment arm based upon an established set of criteria (e.g., a coin toss) is independent of any data collected as a result of previous allocations or participant responses. A common use of this type of fixed allocation strategy is the 'equal allocation' approach where each participant is randomly allocated to either Treatment A or B with equal probability (i.e., $\frac{1}{2}$), regardless of how prior participants were allocated or what their results were. In terms of overall sample size (n) this design is optimal because it will minimise the variance of the estimate of the difference in treatment effects $\hat{\theta}_A - \hat{\theta}_B$ between Treatment Arms A and B when both treatment arms have equal variances and maximise the power of a two-sample t-test [7]. It is this optimal property that provides the basis for why equal allocation is considered a standard reference point against which all adaptive allocation methods are compared.

The theoretical justification for equal allocation follows directly from the variance of the estimated treatment difference. If $\hat{\theta}_A = S_A/n_A$ and $\hat{\theta}_B = S_B/n_B$ are the observed success proportions in each arm, then under independence the variance of $\hat{\theta}_A - \hat{\theta}_B$ is:

$$\text{Var}(\hat{\theta}_A - \hat{\theta}_B) = \frac{\theta_A(1 - \theta_A)}{n_A} + \frac{\theta_B(1 - \theta_B)}{n_B}. \quad (1)$$

For fixed total sample size $n = n_A + n_B$, this expression is minimised by the Neyman allocation $n_A/n = \sqrt{\theta_A(1 - \theta_A)} / [\sqrt{\theta_A(1 - \theta_A)} + \sqrt{\theta_B(1 - \theta_B)}]$, which reduces to $n_A = n_B = n/2$ when $\theta_A = \theta_B$ [7, 4]. Since the true success probabilities are unknown at the design stage, equal allocation serves as a robust default that is optimal under the null hypothesis and near-optimal under moderate departures from it.

In practice, simple equal randomisation does not guarantee an exact split: by chance, one group may have substantially more participants than another, especially if the trial sample size is relatively small. Restrictive randomisation approaches address this problem by placing constraints on the structural organisation of the allocation sequence. The method used most frequently in practice is the Permuted Block Design (PBD). In this method, the sequence is divided into sequential blocks of a predetermined size b . Within each block, $b/2$ participants will be assigned to each arm of the study [7, 6]. This method ensures that there can never be an imbalance greater than $b/2 - 1$ participants at any point during the trial and also that perfect balance is restored after each block.

However, the Permuted block design has some drawbacks. If the size of the blocks is fixed and known to unblinded persons, the assignment of the final participant(s) of each block is non-random, which directly violates allocation concealment and provides the potential for selection bias [7]. Randomly varying block size can reduce, but cannot entirely eliminate this risk. Even more importantly, neither complete randomisation nor the PBD provides an answer to the core ethical question: Both continue to allocate patients to groups in a fixed ratio irrespective of how strong the evidence from the interim analyses suggests one treatment is superior to the other [4]. The main reason for developing adaptive methods to be discussed in the next section is precisely because of these limitations.

For the simulation study presented in this report, the fixed design was implemented as simple complete randomisation with exactly $n/2$ participants per group by using random permutation of a vector of $n/2$ zeros and $n/2$ ones. The use of complete randomisation provided the cleanest possible representation of the theoretical properties of equal allocation and avoided any secondary problems that might arise from the use of block structures. As a result of being implemented in such a manner, the proportion of participants allocated to Group A under this design is fixed at exactly 0.5 in every trial replicate by construction; therefore, the variance of the proportion of participants allocated to Group A across the different simulation replicates is zero. This is an important baseline property: Any increase in the variance of the proportion of participants allocated to Group A observed under adaptive designs reflects the statistical cost associated with responding to the accumulation of outcome data, and this cost is represented in the simulation results of Section 6.

2.3 Adaptive Randomisation: History and Motivation

The origin of adaptive randomisation has been traceable to early research to bridge the gap between the statistical rigor of randomized controlled trials (RCT) and the ethical duties owed to all participants in RCTs. In short, the primary motivation for developing adaptive designs was to reduce or eliminate an ethical cost associated with fixed equal allocation strategies, which continue to assign participants to the inferior treatment(s) even when there is growing evidence of their inferiority; the larger the treatment difference, the greater the ethical costs [4, 8]. As such, adaptive designs have evolved as a systematic response to these concerns by dynamically changing allocation probabilities in response to observed results at each stage of the trial rather than maintaining fixed allocations through out the trial.

The theory of response-adaptive randomised trials was firmly based in a mathematically rigorous model by Hu and Rosenberger [4]. They have shown that an asymptotically valid approach can be developed to assess a wide variety of adaptive allocation strategies. One of their key results relates to the optimum treatment ratio (denoted as ρ^*) which will minimize the expected number of treatment failures while still allowing the estimator of the treatment effect to maintain its required variance level. This optimum ratio is defined for binary outcome measures with success probabilities denoted as θ_A and θ_B , and is given by:

$$\rho^* = \frac{\sqrt{\theta_A(1 - \theta_A)}}{\sqrt{\theta_A(1 - \theta_A)} + \sqrt{\theta_B(1 - \theta_B)}}, \quad (2)$$

Which is reduced to an equal allocation only if the proportions of the two treatments are the same ($\theta_A = \theta_B$). If there is a difference among the treatments, the best allocation is to have as many participants allocated to the better performing treatment as possible; this allocation method also aligns the objective of maximizing the statistical efficiency of the study with the ethical goal of minimizing the number of patients allocated to less effective treatments [4, 7]. Thus, this example illustrates that equal allocation is not always the best choice, and alternative allocations that are statistically justifiable can also be ethically justifiable, provided reliable estimates of the treatment effects can be obtained from the study.

However, since both θ_A and θ_B are unknown and need to be estimated sequentially, this creates an inherent trade-off between exploring the potential of one treatment while exploiting the current advantage of the other treatment. If too much emphasis is placed on favoring the treatment that appears to be producing the better results at any point in time, then too few data may be available for the other treatment, resulting in unstable estimates of its performance and, therefore, uncertainty about how to allocate further participants [4, 8].

The adoption of response-adaptive randomisation in confirmatory trials has nonetheless remained cautious. Because each allocation probability depends on previous outcomes, the assignment sequence is no longer exchangeable, and the standard randomisation-based justification for inference described in Section 2.1 no longer applies directly [1, 7]. Furthermore, because adaptive allocation introduces variability in the realised allocation ratio $\hat{\rho}$, the variance of $\hat{\theta}_A - \hat{\theta}_B$ becomes:

$$\text{Var}(\hat{\theta}_A - \hat{\theta}_B) = \frac{\theta_A(1 - \theta_A)}{n\hat{\rho}} + \frac{\theta_B(1 - \theta_B)}{n(1 - \hat{\rho})}, \quad (3)$$

where $\hat{\rho}$ is now a random variable. The additional variability in $\hat{\rho}$ inflates this expression relative to fixed equal allocation, reducing power at the same total sample size [4]. Simulation-based calibration, as employed in this report, can verify that Type I error control is maintained despite this dependence.

Bayesian Response-Adaptive Methods are a direct extension of the previous Bayesian Response-Adaptive Framework and maintain a full posterior distribution of the success probability of each treatment. Allocation probabilities are computed in a rational manner by using the posterior distribution [4, 8]. By connecting the allocation mechanism directly to the inferential structure of the trial, these methods allow for sequential learning throughout the trial [4, 8]. The model employed here, the Beta-Binomial Conjugate Model with Thompson Sampling, is described in Sections 3.3 and 3.4.

3 Mathematical Foundations

3.1 The Bernoulli and Binomial Models

The statistical framework underpinning this report rests on the Bernoulli and Binomial probability models, which provide the natural and mathematically tractable setting for binary outcome clinical trials [3, 6]. A binary outcome is one in which each participant's response takes one of two values: success (coded as 1) or failure (coded as 0). Let Y_i denote the binary outcome of participant i assigned to a treatment with true success probability $\theta \in (0, 1)$. Then Y_i follows a Bernoulli distribution with probability mass function

$$\mathbb{P}(Y_i = y \mid \theta) = \theta^y(1 - \theta)^{1-y}, \quad y \in \{0, 1\}, \quad (4)$$

mean $\mathbb{E}[Y_i \mid \theta] = \theta$, and variance $\text{Var}(Y_i \mid \theta) = \theta(1 - \theta)$, which achieves its maximum of $\frac{1}{4}$ at $\theta = \frac{1}{2}$. This is precisely the regime of the null hypothesis $H_0 : \theta_A = \theta_B = 0.5$ in Section 6, where uncertainty about treatment outcome is greatest [7].

Assuming participants respond independently, the joint likelihood of an observed sequence $(y_1, \dots, y_n)$ factorises as

$$\mathcal{L}(\theta \mid y_1, \dots, y_n) = \prod_{i=1}^n \theta^{y_i}(1 - \theta)^{1-y_i} = \theta^{S_n}(1 - \theta)^{n-S_n}, \quad (5)$$

where $S_n = \sum_{i=1}^n y_i$ is the total number of observed successes. This demonstrates that S_n is a sufficient statistic for θ , justifying its use as the inferential basis for the hypothesis test in Section 3.2 and the sequential Bayesian updating in Section 3.3 [6]. Since $Y_1, \dots, Y_n$ are independent and identically distributed, S_n follows a Binomial distribution,

$$S_n \mid \theta \sim \text{Binomial}(n, \theta), \quad (6)$$

with probability mass function

$$\mathbb{P}(S_n = s \mid \theta) = \binom{n}{s} \theta^s (1 - \theta)^{n-s}, \quad s \in \{0, 1, \dots, n\}, \quad (7)$$

mean $\mathbb{E}[S_n \mid \theta] = n\theta$, and variance $\text{Var}(S_n \mid \theta) = n\theta(1 - \theta)$. In the two-arm setting, the success counts in each arm satisfy

$$S_A \mid \theta_A \sim \text{Binomial}(n_A, \theta_A), \quad S_B \mid \theta_B \sim \text{Binomial}(n_B, \theta_B), \quad (8)$$

and the natural estimators $\hat{\theta}_A = S_A/n_A$ and $\hat{\theta}_B = S_B/n_B$ are unbiased and consistent. Their difference $\hat{\theta}_A - \hat{\theta}_B$ has sampling variance given by Equation 1 of Section 2.2, which is central to the power analysis of Section 6.2 [7].

The MLE of θ is obtained by maximising the log-likelihood $\ell(\theta) = S_n \log \theta + (n - S_n) \log(1 - \theta)$, yielding $\hat{\theta}_{\text{MLE}} = S_n/n$. The Fisher information for a single Bernoulli observation,

$$\mathcal{I}(\theta) = -\mathbb{E} \left[\frac{\partial^2}{\partial \theta^2} \log \mathbb{P}(Y \mid \theta) \right] = \frac{1}{\theta(1 - \theta)}, \quad (9)$$

implies a Cramér Rao lower bound of $\theta(1 - \theta)/n$. Since $\text{Var}(\hat{\theta}_{\text{MLE}}) = \theta(1 - \theta)/n$, the MLE is a uniformly minimum variance unbiased estimator of θ [7].

3.2 Hypothesis Testing Framework

The primary inferential goal in a two-arm clinical trial is to determine whether the observed difference in treatment outcomes is consistent with a genuine difference in the underlying success probabilities, or whether it can be attributed to chance variation alone. This is formalised through a hypothesis test [3, 6]. Let $\theta_A, \theta_B \in (0, 1)$ denote the true success probabilities of treatments A and B respectively. The null and alternative hypotheses are:

$$H_0 : \theta_A = \theta_B \quad \text{versus} \quad H_1 : \theta_A \neq \theta_B, \quad (10)$$

where the two-sided alternative is adopted throughout, as it places no prior restriction on the direction of any treatment effect. Under H_0 , both arms share a common success probability $\theta_0 = \theta_A = \theta_B$, which must be estimated from the pooled data [7].

The Test Statistic. Given observed success counts S_A and S_B from n_A and n_B participants respectively, the natural estimators of the arm-specific success probabilities are $\hat{\theta}_A = S_A/n_A$ and $\hat{\theta}_B = S_B/n_B$. Under H_0 , the pooled estimator of the common success probability is

$$\hat{\theta}_0 = \frac{S_A + S_B}{n_A + n_B}, \quad (11)$$

which makes efficient use of all available data under the null. The standard two-sample proportions test statistic is then

$$Z = \frac{\hat{\theta}_A - \hat{\theta}_B}{\sqrt{\hat{\theta}_0(1 - \hat{\theta}_0) \left(\frac{1}{n_A} + \frac{1}{n_B} \right)}}, \quad (12)$$

where the denominator is the standard error of $\hat{\theta}_A - \hat{\theta}_B$ evaluated under H_0 . By the central limit theorem, $Z \xrightarrow{d} \mathcal{N}(0, 1)$ as $n_A, n_B \rightarrow \infty$ under H_0 , provided both $n_A \hat{\theta}_0$ and $n_A(1 - \hat{\theta}_0)$ are sufficiently large. This asymptotic normality justifies the use of standard normal critical values for inference [7].

The Rejection Rule. At significance level α , the null hypothesis is rejected whenever

$$|Z| > z_{\alpha/2}, \quad (13)$$

where $z_{\alpha/2}$ denotes the upper $\alpha/2$ quantile of the standard normal distribution. Throughout this report, $\alpha = 0.05$ is adopted as the nominal significance level, giving $z_{0.025} = 1.96$. Equivalently, H_0 is rejected when the two-sided p -value satisfies

$$p = 2(1 - \Phi(|Z|)) < 0.05, \quad (14)$$

where Φ denotes the standard normal cumulative distribution function. In the simulation study, this test is implemented via `prop.test` in R with the argument `correct = FALSE`, suppressing the Yates continuity correction. This is a deliberate choice: the continuity correction is known to be overly conservative in moderate to large samples, and since all simulation scenarios use $n = 100$ or $n = 500$, the uncorrected statistic provides a more accurate approximation to the asymptotic null distribution [6, 7].

Type I Error and Power. Two operating characteristics follow directly from this framework. The Type I error rate is the probability of rejecting H_0 when it is in fact true,

$$\alpha = \mathbb{P}(\text{reject } H_0 \mid H_0 \text{ true}) = \mathbb{P}(|Z| > z_{\alpha/2} \mid \theta_A = \theta_B), \quad (15)$$

and the power of the test against a specific alternative $\theta_A \neq \theta_B$ is

$$1 - \beta = \mathbb{P}(\text{reject } H_0 \mid H_1 \text{ true}) = \mathbb{P}(|Z| > z_{\alpha/2} \mid \theta_A \neq \theta_B). \quad (16)$$

These two quantities capture an inherent tension in trial design: for fixed sample size n , reducing α increases the rejection threshold $z_{\alpha/2}$, which necessarily reduces power. This tension is exacerbated under adaptive randomisation, where the variability in the realised allocation ratio $\hat{\rho}$ inflates the variance of $\hat{\theta}_A - \hat{\theta}_B$ relative to fixed equal allocation, as established in Section 2.3. The formal definitions of all operating characteristics used in the simulation study are given in Section 3.5 [7].

3.3 Bayesian Inference: Beta-Binomial Conjugate Model

In the frequentist framework of Section 3.2, θ is treated as a fixed unknown constant and inference proceeds solely through the observed data. The Bayesian framework instead treats θ as a random variable, encoding uncertainty about its value in a probability distribution that is updated sequentially as participant outcomes accumulate. This is particularly natural in the adaptive randomisation setting, where updated beliefs about θ_A and θ_B directly inform subsequent allocation decisions [4, 8]. The mathematical convenience of this approach is greatly enhanced by the existence of a conjugate prior for the Binomial likelihood, namely the Beta distribution, which ensures the posterior admits a closed-form update rule [4].

The Beta Prior. A random variable $\theta \in (0, 1)$ follows a Beta distribution with shape parameters $\alpha_0, \beta_0 > 0$, written $\theta \sim \text{Beta}(\alpha_0, \beta_0)$, with probability density function

$$\pi(\theta) = \frac{\theta^{\alpha_0-1}(1-\theta)^{\beta_0-1}}{B(\alpha_0, \beta_0)}, \quad \theta \in (0, 1), \quad (17)$$

where $B(\alpha_0, \beta_0) = \int_0^1 t^{\alpha_0-1}(1-t)^{\beta_0-1} dt$ is the normalising Beta function, with mean $\mathbb{E}[\theta] = \alpha_0/(\alpha_0 + \beta_0)$ and variance $\text{Var}(\theta) = \alpha_0\beta_0/[(\alpha_0 + \beta_0)^2(\alpha_0 + \beta_0 + 1)]$. The hyperparameters jointly control the location and concentration of the prior, with large values of $\alpha_0 + \beta_0$ encoding strong prior beliefs and small values reflecting greater uncertainty. The non-informative prior $\text{Beta}(1, 1)$, reducing to the uniform distribution on $(0, 1)$, is adopted for both arms throughout, reflecting complete prior ignorance about each treatment's success probability before the trial begins [8].

The Posterior Distribution. After observing S_n successes from n participants, Bayes' theorem gives the posterior as proportional to the product of the Binomial likelihood and the Beta prior,

$$\begin{aligned} \pi(\theta | S_n) &\propto \theta^{S_n}(1-\theta)^{n-S_n} \cdot \theta^{\alpha_0-1}(1-\theta)^{\beta_0-1} \\ &= \theta^{(\alpha_0+S_n)-1}(1-\theta)^{(\beta_0+n-S_n)-1}, \end{aligned} \quad (18)$$

which is recognisable as the kernel of a Beta distribution, yielding the conjugate posterior

$$\theta | S_n \sim \text{Beta}(\alpha_0 + S_n, \beta_0 + n - S_n). \quad (19)$$

Each observed success increments α by one and each failure increments β by one. The posterior mean

$$\mathbb{E}[\theta | S_n] = \frac{\alpha_0 + \beta_0}{\alpha_0 + \beta_0 + n} \cdot \frac{\alpha_0}{\alpha_0 + \beta_0} + \frac{n}{\alpha_0 + \beta_0 + n} \cdot \frac{S_n}{n} \quad (20)$$

is a convex combination of the prior mean and the MLE S_n/n , shrinking towards the prior when n is small and towards the MLE as $n \rightarrow \infty$, reflecting the asymptotic equivalence of Bayesian and frequentist inference under regular models [4].

Sequential Updating. The conjugate structure ensures the posterior after j participants serves as the prior for participant $j+1$. If the current posterior is $\text{Beta}(\alpha_j, \beta_j)$, then after observing $y_{j+1} \in \{0, 1\}$,

$$\alpha_{j+1} = \alpha_j + y_{j+1}, \quad \beta_{j+1} = \beta_j + (1 - y_{j+1}), \quad (21)$$

summarising the entire inferential history in just two scalars at each stage. In the two-arm setting, independent models are maintained for each arm,

$$\theta_A | S_A \sim \text{Beta}(\alpha_0 + S_A, \beta_0 + n_A - S_A), \quad \theta_B | S_B \sim \text{Beta}(\alpha_0 + S_B, \beta_0 + n_B - S_B), \quad (22)$$

with both posteriors updated after every allocation. This sequential learning mechanism is the inferential foundation upon which Thompson Sampling in Section 3.4 constructs its adaptive allocation rule [4, 8].

3.4 Thompson Sampling

Thompson Sampling is a Bayesian adaptive allocation strategy that resolves the explore-exploit tension inherent in sequential decision making by randomising allocation probabilities in proportion to the posterior probability that each treatment is superior. Rather than deterministically assigning all participants to the arm with the highest posterior mean, Thompson Sampling preserves uncertainty-driven exploration whilst progressively favouring the arm that accumulates stronger evidence of superiority. This makes it particularly well-suited to the clinical trial setting, where both statistical validity and ethical allocation must be maintained simultaneously [4, 8].

The Allocation Rule. At each stage j of the trial, independent samples are drawn from the current posterior distributions of each arm,

$$\tilde{\theta}_A^{(j)} \sim \text{Beta}(\alpha_0 + S_A^{(j)}, \beta_0 + n_A^{(j)} - S_A^{(j)}), \quad \tilde{\theta}_B^{(j)} \sim \text{Beta}(\alpha_0 + S_B^{(j)}, \beta_0 + n_B^{(j)} - S_B^{(j)}), \quad (23)$$

where $S_A^{(j)}$, $n_A^{(j)}$ and $S_B^{(j)}$, $n_B^{(j)}$ denote the cumulative successes and participant counts in each arm up to stage j . Participant $j + 1$ is then allocated to arm A if $\tilde{\theta}_A^{(j)} > \tilde{\theta}_B^{(j)}$, and to arm B otherwise. The resulting probability of allocation to arm A is

$$p_A^{(j)} = \mathbb{P}(\tilde{\theta}_A^{(j)} > \tilde{\theta}_B^{(j)}) = \mathbb{P}(\theta_A > \theta_B \mid \text{data up to stage } j), \quad (24)$$

which is precisely the posterior probability that treatment A is superior given all outcomes observed so far. This quantity increases towards one as evidence in favour of arm A accumulates, and decreases towards zero as evidence favours arm B , providing a natural and coherent mechanism for adaptive allocation [4].

Monte Carlo Approximation. The posterior probability $p_A^{(j)}$ has no closed-form expression, as it requires evaluating

$$p_A^{(j)} = \int_0^1 F_B(\theta; \alpha_0 + S_B^{(j)}, \beta_0 + n_B^{(j)} - S_B^{(j)}) f_A(\theta; \alpha_0 + S_A^{(j)}, \beta_0 + n_A^{(j)} - S_A^{(j)}) d\theta, \quad (25)$$

where f_A and F_B denote the Beta density and cumulative distribution function of arms A and B respectively. This integral is approximated at each stage by drawing M independent samples from each posterior and computing

$$\hat{p}_A^{(j)} = \frac{1}{M} \sum_{m=1}^M \mathbf{1}(\tilde{\theta}_A^{(j,m)} > \tilde{\theta}_B^{(j,m)}), \quad (26)$$

where $\tilde{\theta}_A^{(j,m)}$ and $\tilde{\theta}_B^{(j,m)}$ are the m -th independent draws from the respective posteriors. By the law of large numbers, $\hat{p}_A^{(j)} \rightarrow p_A^{(j)}$ almost surely as $M \rightarrow \infty$. In the simulation study, $M = 500$ posterior samples are used at each allocation step, providing a sufficiently precise approximation whilst remaining computationally efficient [8].

Clipping and Burn-in. Two practical constraints are imposed on the allocation probability to ensure statistical validity throughout the trial. First, $\hat{p}_A^{(j)}$ is clipped to the interval $[\epsilon, 1 - \epsilon]$ with $\epsilon = 0.1$, so that

$$p_A^{(j)} \leftarrow \max\left(\epsilon, \min\left(1 - \epsilon, \hat{p}_A^{(j)}\right)\right), \quad (27)$$

ensuring that neither arm is ever completely abandoned during the trial. Without this constraint, early data could drive $\hat{p}_A^{(j)}$ to near zero or one, leaving one arm with insufficient data to support valid inference at the end of the trial. Second, a burn-in period of $b = 10$ participants is implemented at the start of the trial, during which equal allocation $p_A^{(j)} = 0.5$ is enforced regardless of the data. This ensures that both posteriors are sufficiently informed before Thompson Sampling begins, preventing highly unstable allocation probabilities arising from the uninformative $\text{Beta}(1, 1)$ prior in the earliest stages of the trial [8].

3.5 Operating Characteristics: Formal Definitions

The operating characteristics of a trial design are the quantitative criteria by which its statistical and ethical performance are evaluated. Three operating characteristics are used to compare fixed and adaptive randomisation throughout this report: the Type I error rate, statistical power, and the proportion of participants allocated to the superior treatment. Together these capture the fundamental tension between statistical efficiency and ethical allocation that is central to this report [7].

Type I Error Rate. The Type I error rate is the probability of incorrectly rejecting H_0 when no true treatment difference exists,

$$\alpha = \mathbb{P}(\text{reject } H_0 \mid \theta_A = \theta_B) = \mathbb{E}[\mathbf{1}(p\text{-value} < 0.05 \mid \theta_A = \theta_B)], \quad (28)$$

where the expectation is taken over the randomisation distribution. A design is said to maintain nominal Type I error control if $\alpha \leq 0.05$, which is a minimum requirement for any valid inferential procedure. Under adaptive randomisation this is less immediate, since the dependence introduced by the adaptive allocation mechanism means the standard randomisation-based justification for inference no longer applies directly, as noted in Section 2.3 [7].

Statistical Power. Let $\delta = \theta_A - \theta_B$ denote the true treatment difference. The power of the test against a specific alternative is

$$\pi(\delta) = \mathbb{P}(\text{reject } H_0 \mid \theta_A - \theta_B = \delta), \quad (29)$$

where $\pi(\delta) \rightarrow \alpha$ as $\delta \rightarrow 0$ and $\pi(\delta) \rightarrow 1$ as $|\delta| \rightarrow \infty$. The conventional target of $\pi(\delta) \geq 0.80$ is adopted throughout. As established in Section 2.3, adaptive randomisation introduces variability in the realised allocation ratio $\hat{\rho}$, inflating $\text{Var}(\hat{\theta}_A - \hat{\theta}_B)$ and reducing power at fixed total sample size n [7].

Ethical Allocation. Let $\hat{\rho} = n_A/n$ denote the realised allocation proportion to arm A , and suppose without loss of generality that $\theta_A > \theta_B$. The ethical allocation characteristic is

$$\eta = \mathbb{E}[\hat{\rho} \mid \theta_A > \theta_B], \quad (30)$$

with $\eta = 0.5$ under fixed equal allocation by construction and $\eta > 0.5$ under any response-adaptive design that successfully skews allocation towards the superior arm. The theoretical upper bound on η is the optimal Neyman allocation ratio ρ^* defined in Section 2.3 [4].

Monte Carlo Estimation. Let $R_b \in \{0, 1\}$ indicate rejection in replicate b and $\hat{\rho}_b$ denote the realised allocation proportion in replicate b . The Monte Carlo estimators are

$$\hat{\alpha} = \frac{1}{B} \sum_{b=1}^B R_b, \quad \hat{\pi} = \frac{1}{B} \sum_{b=1}^B R_b, \quad \hat{\eta} = \frac{1}{B} \sum_{b=1}^B \hat{\rho}_b, \quad (31)$$

evaluated under H_0 , H_1 and H_1 respectively. Since $\hat{\alpha}$ and $\hat{\pi}$ are sample proportions, their Monte Carlo standard errors are

$$\text{SE}(\hat{\alpha}) = \sqrt{\frac{\hat{\alpha}(1 - \hat{\alpha})}{B}}, \quad \text{SE}(\hat{\pi}) = \sqrt{\frac{\hat{\pi}(1 - \hat{\pi})}{B}}, \quad (32)$$

which are bounded above by $1/(2\sqrt{B})$. With $B = 2000$ replicates this gives an upper bound of approximately 0.011, ensuring all estimated operating characteristics are precise to within one percentage point [7, 8].

4 Randomisation Methods

4.1 Fixed 1:1 Randomisation

Under fixed 1:1 randomisation, each of the n participants is allocated to treatment arm A or arm B with equal probability $\frac{1}{2}$, independently of all previous allocations and observed outcomes [7]. The allocation sequence is generated by a random permutation of a vector containing exactly $n/2$ ones and $n/2$ zeros, ensuring perfect balance by construction. As established in Section 2.2, and consistent with standard practice in clinical trial design [6, 7], this design minimises $\text{Var}(\hat{\theta}_A - \hat{\theta}_B)$ under the null hypothesis and serves as the natural reference point against which all adaptive designs are compared throughout this report.

The key statistical property of this design is that the realised allocation proportion $\hat{\rho} = n_A/n = 0.5$ is fixed in every replicate by construction, so the variance of $\hat{\rho}$ across replicates is identically zero. This is an important baseline: as shown in Section 2.3, any increase in $\text{Var}(\hat{\rho})$ observed under adaptive designs directly inflates $\text{Var}(\hat{\theta}_A - \hat{\theta}_B)$ and represents the statistical cost of responding to accumulating outcome data [4, 7]. Fixed 1:1 randomisation incurs no such cost, which is precisely why it maximises power at fixed total sample size n and remains the default design in confirmatory trials [7]. Its principal limitation, however, is that it continues to allocate participants to the inferior treatment at a fixed rate of 50% regardless of how strong the interim evidence for treatment superiority becomes, raising the ethical concerns discussed in Section 2.3 [4, 8].

4.2 Simple Response-Adaptive Randomisation

Simple Response-Adaptive Randomisation adjusts the allocation probability after each participant based on the observed success rates in each arm. Let $\hat{r}_A^{(j)}$ and $\hat{r}_B^{(j)}$ denote the Laplace-smoothed success rates in each arm after j participants,

$$\hat{r}_A^{(j)} = \frac{S_A^{(j)} + 0.5}{n_A^{(j)} + 1}, \quad \hat{r}_B^{(j)} = \frac{S_B^{(j)} + 0.5}{n_B^{(j)} + 1}, \quad (33)$$

where the additive smoothing prevents division by zero in the earliest stages of the trial when one or both arms may have received very few participants. The allocation probability for participant $j + 1$ is then set to:

$$p_A^{(j)} = \begin{cases} p_{\text{up}} & \text{if } \hat{r}_A^{(j)} > \hat{r}_B^{(j)}, \\ p_{\text{down}} & \text{if } \hat{r}_B^{(j)} > \hat{r}_A^{(j)}, \\ 0.5 & \text{if } \hat{r}_A^{(j)} = \hat{r}_B^{(j)}, \end{cases} \quad (34)$$

with $p_{\text{up}} = 0.6$ and $p_{\text{down}} = 0.4$ throughout. This rule favours the currently superior arm whilst preserving a non-trivial probability of allocation to the inferior arm, maintaining the randomisation necessary for valid inference at the end of the trial [4, 7].

The simplicity of this approach is both its strength and its limitation. Unlike the Bayesian approach of Section 4.3, simple RAR does not maintain a full probability model over θ_A and θ_B ; instead it responds directly to point estimates of treatment performance, making it computationally straightforward but statistically less principled [4, 8]. In particular, the allocation rule is binary in nature: regardless of how large the observed difference in success rates is, the allocation probability can only take one of three values $\{0.4, 0.5, 0.6\}$, which limits the degree to which the design can skew allocation towards the superior arm even when the evidence strongly favours one treatment. This motivates the Bayesian approach, which allows the allocation probability to vary continuously in response to the full posterior distribution of each arm's success probability [4, 8].

4.3 Bayesian Response-Adaptive Randomisation

Bayesian Response-Adaptive Randomisation combines the Beta-Binomial conjugate model of Section 3.3 with the Thompson Sampling allocation rule of Section 3.4 to produce a fully probabilistic adaptive design. Unlike simple RAR, which responds only to point estimates of treatment performance, Bayesian RAR maintains a full posterior distribution over each arm’s success probability and uses the posterior probability of superiority to determine the allocation at each stage. This allows the allocation probability to vary continuously throughout the trial in a manner that is directly calibrated to the accumulated evidence, rather than being restricted to a fixed set of values [4, 8].

The Sequential Algorithm. The Bayesian RAR procedure operates as follows. Prior to the trial, independent Beta priors are assigned to each arm,

$$\theta_A \sim \text{Beta}(\alpha_0, \beta_0), \quad \theta_B \sim \text{Beta}(\alpha_0, \beta_0), \quad (35)$$

with $\alpha_0 = \beta_0 = 1$ throughout, reflecting complete prior ignorance. For the first $b = 10$ participants, equal allocation $p_A^{(j)} = 0.5$ is enforced as a burn-in period to ensure both posteriors are sufficiently informed before adaptive allocation begins. For participant $j > b$, the allocation probability is computed as the posterior probability that arm A is superior,

$$p_A^{(j)} = \mathbb{P}(\theta_A > \theta_B \mid \text{data up to stage } j), \quad (36)$$

approximated via Monte Carlo using $M = 500$ posterior samples as described in Section 3.4, and clipped to the interval $[0.1, 0.9]$ to ensure neither arm is completely abandoned. Participant j is then allocated to arm A with probability $p_A^{(j)}$ and to arm B with probability $1 - p_A^{(j)}$, and the posterior of the assigned arm is updated according to the conjugate rule of Section 3.3,

$$\alpha_{j+1} = \alpha_j + y_j, \quad \beta_{j+1} = \beta_j + (1 - y_j), \quad (37)$$

where $y_j \in \{0, 1\}$ is the observed outcome of participant j . This three-step cycle of sampling, allocating, and updating repeats for every participant throughout the trial [4, 8].

Key Properties. Several properties distinguish Bayesian RAR from the simpler adaptive approach of Section 4.2. First, the allocation probability $p_A^{(j)}$ varies continuously on the clipped interval $[0.1, 0.9]$ rather than taking only three discrete values, allowing the design to respond proportionally to the strength of evidence rather than merely its direction. Second, the allocation rule is directly connected to the inferential structure of the trial: $p_A^{(j)}$ is not an ad hoc rule but a coherent Bayesian quantity that has a precise probabilistic interpretation at every stage. Third, as the sample size grows and the posteriors concentrate around the true values θ_A and θ_B , the allocation probability converges towards

$$p_A^{(j)} \longrightarrow \mathbf{1}(\theta_A > \theta_B) \quad \text{as } j \rightarrow \infty, \quad (38)$$

so that in the limit all participants are allocated to the superior arm. In finite samples, the clipping constraint ensures this convergence is tempered, preserving sufficient allocation to both arms for valid inference. Furthermore, as noted in Section 2.3, the expected allocation ratio under Bayesian RAR approximates the optimal Neyman ratio ρ^* , aligning the ethical goal of minimising treatment failures with the statistical goal of minimising $\text{Var}(\hat{\theta}_A - \hat{\theta}_B)$ [4].

The principal cost of this design relative to fixed 1:1 randomisation is a reduction in statistical power at fixed total sample size n . Because $\hat{\rho}$ is now a random variable rather than a fixed constant, $\text{Var}(\hat{\theta}_A - \hat{\theta}_B)$ is inflated relative to the fixed design, as established in Equation 3 of Section 2.3 [7]. The extent of this power loss depends on the magnitude of the true treatment difference $\delta = \theta_A - \theta_B$: when δ is large, the allocation skews heavily towards arm A , increasing $\text{Var}(\hat{\rho})$ and amplifying the power loss. This represents the fundamental statistical-ethical trade-off that is the central focus of this report, and its quantification via Monte Carlo simulation is the primary objective of Section 6 [4, 7, 8].

4.4 Cluster Randomisation

In individually randomised trials, the unit of randomisation and the unit of analysis coincide: each participant is independently assigned to a treatment arm. In some settings, however, this is either impractical or inappropriate. When treatments operate at the group level, such as a public health intervention delivered to an entire community, or when there is a risk of contamination between participants in close contact, it is necessary to randomise entire clusters rather than individuals [2, 10]. In a cluster randomised trial, each cluster $c = 1, \dots, C$ is assigned as a whole to either arm A or arm B , and all k participants within the cluster receive the same treatment. The statistical consequence of this design is that outcomes within the same cluster are no longer independent, violating the core assumption underlying the individual-level analysis of Sections 3.1 and 3.2 [6].

The Intracluster Correlation Coefficient. The degree of outcome similarity within clusters is quantified by the intracluster correlation coefficient (ICC), defined as

$$\rho = \frac{\text{Var}(\mu_c)}{\text{Var}(\mu_c) + \text{Var}(Y_{ic} | \mu_c)}, \quad (39)$$

where μ_c denotes the true success probability of cluster c and Y_{ic} denotes the binary outcome of participant i within cluster c . The ICC takes values in $[0, 1]$: $\rho = 0$ corresponds to complete independence between participants within a cluster, recovering the individually randomised setting, whilst $\rho = 1$ corresponds to perfect within-cluster homogeneity, where all participants in a cluster share the same outcome. In practice, ICC values in clinical and public health trials typically lie in the range $[0.01, 0.20]$, though even small values can substantially reduce the effective sample size of the clinical trial [2, 10].

The Design Effect. The design effect quantifies the inflation in variance arising from cluster randomisation relative to individual randomisation with the same total number of participants. For clusters of equal size k , the design effect is

$$\text{DEFF} = 1 + (k - 1)\rho, \quad (40)$$

so that the effective sample size of a cluster randomised trial with C clusters per arm and k participants per cluster is

$$n_{\text{eff}} = \frac{Ck}{\text{DEFF}} = \frac{Ck}{1 + (k - 1)\rho}. \quad (41)$$

This expression makes explicit the cost of clustering: for fixed total sample size $n = Ck$, increasing either k or ρ reduces n_{eff} and therefore reduces power. For example, with $k = 10$ participants per cluster and $\rho = 0.05$, the design effect is $\text{DEFF} = 1.45$, meaning the trial has the same power as an individually randomised trial with only $n/1.45 \approx 69\%$ of the total participants [2, 10].

The Beta-Binomial Cluster Model. To model between-cluster variability in success probabilities, the cluster-level success probability μ_c is assigned a Beta distribution,

$$\mu_c \sim \text{Beta}(a, b), \quad a = p\phi, \quad b = (1 - p)\phi, \quad \phi = \frac{1 - \rho}{\rho}, \quad (42)$$

where p is the marginal success probability and ϕ is a precision parameter determined entirely by the ICC ρ . This parametrisation ensures that $\mathbb{E}[\mu_c] = p$ and $\text{Var}(\mu_c) = p(1 - p)\rho$, so the ICC is directly encoded in the variance of the cluster-level success probabilities. Conditional on μ_c , the outcomes of the k participants within cluster c are independent Bernoulli trials with success probability μ_c , yielding a Beta-Binomial marginal distribution for the cluster success count that naturally accommodates overdispersion relative to the Binomial model of Section 3.1 [2].

Inference. Since the unit of randomisation is the cluster rather than the individual, inference proceeds on cluster-level proportions rather than individual outcomes. Let $\bar{Y}_c = k^{-1} \sum_{i=1}^k Y_{ic}$ denote the observed success proportion in cluster c . Under the null hypothesis of no treatment effect, a two-sample Welch t -test is applied to the cluster proportions $\{\bar{Y}_c\}_{c \in A}$ and $\{\bar{Y}_c\}_{c \in B}$, which accounts for unequal within-arm variances without requiring parametric assumptions about the distribution of μ_c . This replaces the individual-level proportions test of Section 3.2, which would be invalid here due to the within-cluster dependence induced by the clustering structure [2, 10].

5 Simulation Study Design

5.1 Data generating Mechanism

This simulation study evaluates the operating characteristics of the three individual-level randomisation designs: Fixed 1:1, Simple RAR, and Bayesian RAR, under a fully specified and transparent data generating mechanism. In every simulated trial, n participants are sequentially enrolled and assigned to either Treatment Arm A or Treatment Arm B according to the allocation rule corresponding to each design. Each participant's binary outcome is then drawn independently from a Bernoulli distribution, with the success probability equal to the treatment received.

$$Y_i \mid (\text{arm} = A) \sim \text{Bernoulli}(\theta_A), \quad Y_i \mid (\text{arm} = B) \sim \text{Bernoulli}(\theta_B), \quad (43)$$

In this equation, θ_A and θ_B represent the actual probabilities of success for Arms A and B. The parametrisation follows that of the Bernoulli model in Section 3.1 meaning total consistency between theory and the simulation. The generating mechanism for the data is thus completely characterised by the pair (θ_A, θ_B) , with no nuisance parameters and no model misspecification [3, 6].

Two simulation scenarios are taken as a reference throughout the study. In the null scenario, with $\theta_A = \theta_B = 0.5$, we assume there is no true treatment difference and the relevant operating characteristic is the Type I error rate. The values $\theta_A = \theta_B = 0.5$ were chosen because at this value Bernoulli outcomes have maximum variance – $\text{Var}(Y_i) = \theta(1 - \theta)$ achieves maximum value of $\frac{1}{4}$ at $\theta = \frac{1}{2}$ [7]. This presents the most difficult challenge for control of Type I error across all possible null configurations. In the other scenario, θ_B is fixed at 0.50 throughout, while θ_A varies at the values $\{0.55, 0.65, 0.80\}$ which corresponds to small, moderate, and large treatment effect sizes respectively. The Hu and Rosenberger theoretical results can be directly compared to this by choosing the baseline probability $\theta_B = 0.5$, which is a standard choice in the comparative adaptive randomisation literature [4].

The major sample size under consideration throughout the simulation study is $n = 100$, which is a representative count of participants in a phase II clinical trial where an adaptive design for the study is most often under consideration [3, 6]. In Section 5.4, a sensitivity analysis at $n = 500$ is conducted to determine the robustness of the results to this choice. At all times the total sample size n is held constant across designs, allowing power and ethical allocation comparisons to be made on a like-for-like basis.

5.2 R Implementation

All simulations were implemented in R through custom functions written from scratch for the present study (code can be found in Appendix A). A random seed is fixed through `set.seed(123)` for complete reproducibility. The three individual-level designs require separate functions: `run_fixed_sim`, `run_rar_sim`, and `run_brar_sim`. There are also functions for the cluster and covariate-adjusted extensions in Sections 7 and 8. Within each replicate, participants are enrolled sequentially and assigned according to the allocation rule of the design being simulated. Outcomes are drawn from the appropriate Bernoulli distribution. At the end of each replicate, $H_0 : \theta_A = \theta_B$ is tested via `prop.test` with `correct = FALSE`, and the rejection indicator, the realised allocation proportion $\hat{\rho}$, and the total successes are recorded. For Bayesian RAR, the posterior probability of superiority is approximated using $M = 500$ posterior samples per allocation step, with burn-in $b = 10$ and clipping to $[0.1, 0.9]$, as described in Section 3.4. The principal suite performs all three designs across all scenarios at $n = 100$ and $n = 500$, yielding 48,000 individual trial simulations at $B = 2000$ replicates each [3, 7, 8].

5.3 Monte Carlo Error and Precision

It is essential to quantify the precision of all estimates, as all operating characteristics are estimated by Monte Carlo simulation. According to Section 3.5, the Monte Carlo standard error associated with an estimate $\hat{p}$ based on B replicates is given by:

$$\text{SE}(\hat{p}) = \sqrt{\frac{\hat{p}(1 - \hat{p})}{B}}, \quad (44)$$

and is bounded above by $1/(2\sqrt{B})$ regardless of p . With $B = 2000$ replicates, this upper bound is approximately 0.011, thus ensuring that all estimated Type I error rates and power values mentioned in Section 6 are accurate to within one percent.

This precision is illustrated in Figure 9, which plots the standard error of $\hat{p}$ as a function of p for $B \in \{500, 1000, 2000\}$. The figure confirms that $B = 2000$ keeps the Monte Carlo standard error strictly below 0.011 across the entire range of $p \in (0, 1)$, whilst $B = 500$ can yield standard errors approaching 0.022 near $p = 0.5$. As a result, all numerical outputs in Section 6 are provided with 95% Monte Carlo confidence intervals, computed as $\hat{p} \pm 1.96 \cdot \text{SE}(\hat{p})$, throughout.

For the ethical allocation characteristic $\hat{\eta} = \mathbb{E}[\hat{\rho}]$, the Monte Carlo standard error is estimated as the standard deviation of the B realised allocation proportions $\hat{\rho}_1, \dots, \hat{\rho}_B$ divided by $\sqrt{B}$. Because Fixed 1:1 randomisation results in $\hat{\rho} = 0.5$ in every replicate by construction, this standard error is identically zero and is not reported for that design. For adaptive designs, the standard deviation of $\hat{\rho}$ across replicates quantifies the allocation variability leading to the power reduction shown in Section 2.3 [7, 8].

5.4 Sensitivity Analysis

The first simulation study utilises a sample size of $n = 100$. To examine the sensitivity of the conclusions to this choice, the complete simulation suite is repeated at $n = 500$ in Section 6.5. This larger sample size is more representative of Phase III confirmatory trials and allows examination of whether the power advantage of Fixed 1:1 randomisation over the adaptive designs diminishes with increasing n , as can theoretically be predicted from the consistency of the three estimators and the fact that the adaptive allocation ratios converge to their optimal values as n increases [3, 7]. It should be noted that as n increases, the variance of the realised allocation proportion $\hat{\rho}$ under adaptive designs decreases, which reduces the variance inflation identified in Section 2.3 and narrows the power gap between fixed and adaptive designs [4, 7].

For the cluster randomisation extension of Section 7, the ICC ρ is varied over $\{0.00, 0.02, 0.05, 0.10, 0.20\}$, covering the range of values typically encountered in clinical and public health cluster randomised trials [2, 10]. This allows direct examination of how the design effect $\text{DEFF} = 1 + (k - 1)\rho$ grows with ρ and how this translates into power loss at fixed total sample size $n = 200$, corresponding to 20 clusters per arm with $k = 10$ participants per cluster. The ICC value of $\rho = 0.00$ represents the baseline individually randomised setting, from which the cost of clustering can be directly measured as ρ increases [10].

In the covariate-adjusted RAR extension of Section 8, the covariate effect size β_x is varied across $\{0.0, 0.3, 0.6, 1.0, 1.5\}$ on the log-odds scale. This addresses the practical question of the strength of association between the binary covariate X and the outcome that is required for covariate adjustment to confer a meaningful improvement in power or ethical allocation over the unadjusted designs. When $\beta_x = 0$ there is no covariate effect and CA-RAR reduces to standard Bayesian RAR, providing a natural internal consistency check for the simulation [8, 10].

6 Results

6.1 Type I Error Control

Table 1 displays all estimated operating characteristics at $n = 100$ across all scenarios and methods. Under the null scenario where $\theta_A = \theta_B = 0.50$, all three designs maintain adequate Type I error control. The Type I error rate of Fixed 1:1 and Simple RAR is 0.056, which is slightly above the nominal level of 0.05 but within the bounds of Monte Carlo sampling variation — the 95% confidence intervals of (0.047, 0.068) and (0.046, 0.067) both include 0.05. Bayesian RAR produces a rate of 0.061, with confidence interval (0.051, 0.073), which also includes 0.05. As such, none of the three designs exhibit systematic inflation of the Type I error rate at the nominal significance level of $\alpha = 0.05$. This finding is noteworthy since adaptive designs may inflate Type I error due to the data-dependent allocation mechanism, as discussed in Section 2.3 [8]. The nominal control of Bayesian RAR confirms that the burn-in period and clipping constraint of Section 3.4 are effective in preserving the statistical validity of the trial [8].

Table 1: Estimated operating characteristics at $n = 100$, $\theta_B = 0.50$, $B = 2000$ replicates. Power/Type I reports power under H_1 and Type I error rate under H_0 . CI denotes 95% Monte Carlo confidence interval. Alloc SD is not defined for Fixed 1:1 by construction.

Scenario	Method	Power/Type I	MC SE	95% CI	Alloc to A	Alloc SD
$\theta_A = 0.50$ (null)	Fixed 1:1	0.056	0.0052	(0.047, 0.068)	0.500	–
$\theta_A = 0.50$ (null)	Simple RAR	0.056	0.0051	(0.046, 0.067)	0.499	0.087
$\theta_A = 0.50$ (null)	Bayesian RAR	0.061	0.0054	(0.051, 0.073)	0.493	0.231
$\theta_A = 0.55$ (small)	Fixed 1:1	0.083	0.0062	(0.071, 0.096)	0.500	–
$\theta_A = 0.55$ (small)	Simple RAR	0.071	0.0057	(0.060, 0.083)	0.526	0.083
$\theta_A = 0.55$ (small)	Bayesian RAR	0.092	0.0065	(0.080, 0.106)	0.571	0.225
$\theta_A = 0.65$ (moderate)	Fixed 1:1	0.331	0.0105	(0.310, 0.352)	0.500	–
$\theta_A = 0.65$ (moderate)	Simple RAR	0.350	0.0107	(0.330, 0.372)	0.563	0.069
$\theta_A = 0.65$ (moderate)	Bayesian RAR	0.272	0.0100	(0.253, 0.292)	0.721	0.173
$\theta_A = 0.80$ (large)	Fixed 1:1	0.895	0.0068	(0.881, 0.908)	0.500	–
$\theta_A = 0.80$ (large)	Simple RAR	0.893	0.0069	(0.878, 0.906)	0.588	0.053
$\theta_A = 0.80$ (large)	Bayesian RAR	0.745	0.0097	(0.725, 0.764)	0.826	0.070

6.2 Statistical Power

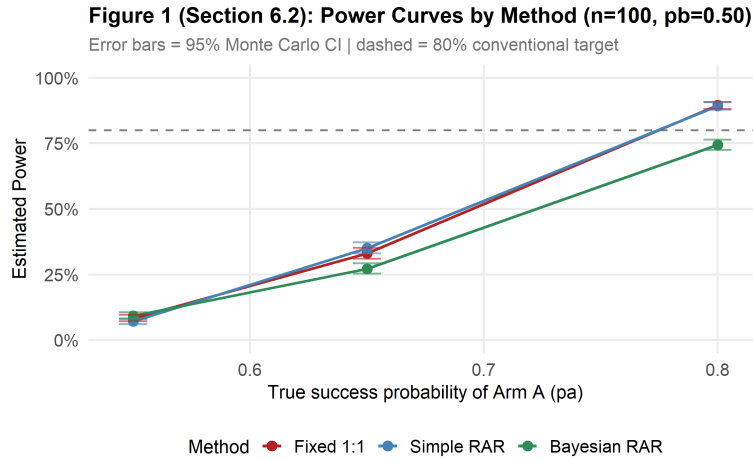


Figure 1: Power curves for Fixed 1:1, Simple RAR, and Bayesian RAR across alternative scenarios at $n = 100$, $\theta_B = 0.50$. Error bars denote 95% Monte Carlo confidence intervals. The dashed horizontal line indicates the conventional 80% power target.

Table 1 and Figure 1 present the estimated power of each method across the three alternative scenarios at $n = 100$.

When the difference between the treatments is small ($\theta_A = 0.55$, $\delta = 0.05$), all three methods exhibit low power, as expected given the small sample size and modest treatment difference. Fixed 1:1 achieves 0.083 (95% CI: 0.071, 0.096), Simple RAR 0.071 (0.060, 0.083), and Bayesian RAR 0.092 (0.080, 0.106). All confidence intervals overlap extensively and none of the methods fulfil the 80% power target. With this minor effect size, adaptation does not sway allocation very much — Bayesian RAR sends only 57.1% of patients on average to the better arm — so the increase in variance described in Equation 3 is negligible and the power difference between methods is essentially zero [4, 7].

When the treatments differ moderately ($\theta_A = 0.65$, $\delta = 0.15$), a noticeable gap begins to emerge. Fixed 1:1 achieves 0.331 (95% CI: 0.310, 0.352) and Simple RAR 0.350 (0.330, 0.372), with overlapping confidence intervals indicating the two are essentially indistinguishable. Bayesian RAR achieves a lower 0.272 (0.253, 0.292), whose confidence interval does not overlap with Fixed 1:1 at all. This drop of approximately 6 percentage points occurs because Bayesian RAR sends on average 72.1% of patients to arm A with a standard deviation of $\hat{p}$ of 0.173, increasing the variance of the estimated treatment difference and reducing power as established in Section 2.3 [4, 7].

The gap is largest when the treatment difference is large ($\theta_A = 0.80$, $\delta = 0.30$). Fixed 1:1 achieves 0.895 (95% CI: 0.881, 0.908) and Simple RAR an essentially identical 0.893 (0.878, 0.906), both comfortably exceeding the 80% target. Bayesian RAR achieves only 0.745 (0.725, 0.764), a reduction of approximately 15 percentage points whose confidence interval does not overlap with either of the other two designs, and is the only method that fails to meet the 80% target. This occurs because Bayesian RAR sends 82.6% of patients to arm A, with the posterior probability of arm A superiority saturating at the clip ceiling of 0.90 for the majority of the trial, as illustrated in Figure 3 [8]. This leaves very little data from arm B, increasing the variance of the treatment difference estimate and weakening the test [7].

In essence, these findings support the prediction of Section 2.3: as the treatment difference grows, the power cost of Bayesian RAR increases, driven by progressively heavier allocation skew towards the superior arm [4]. Simple RAR incurs essentially no power cost relative to Fixed 1:1, since its allocation rule only ever goes as far as 0.6 in favour of the better arm [4, 7]. As will be demonstrated in Section 6.3, however, this conservatism also means Simple RAR does not achieve much ethical benefit either [4].

6.3 Ethical Allocation

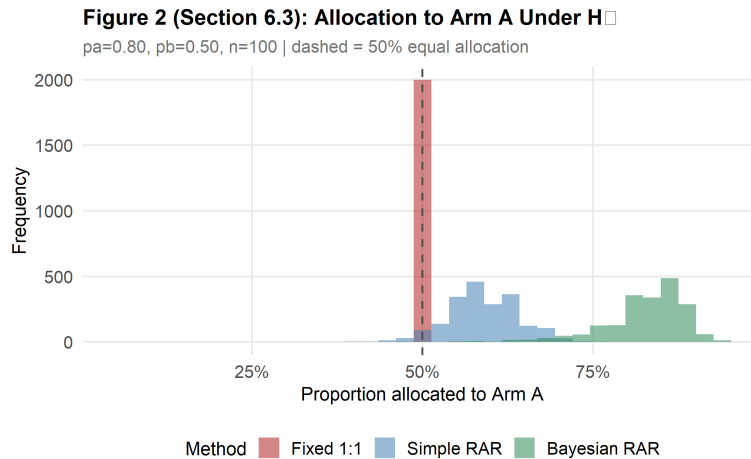


Figure 2: Distribution of the realised allocation proportion $\hat{p}$ to arm A across $B = 2000$ simulation replicates under the large effect scenario ($\theta_A = 0.80$, $\theta_B = 0.50$, $n = 100$). The dashed vertical line indicates 50% equal allocation.

Table 1 and Figure 2 show how ethical each design is in the allocation it gives to the superior arm of the trial, measured by the mean realised allocation proportion as defined in Equation 30. In the null scenario ($\theta_A = \theta_B = 0.50$), all three designs allocate approximately equally — Fixed 1:1 at exactly 0.500, Simple RAR at 0.499, and Bayesian RAR at 0.493. Thus, neither adaptive design has any systematic bias in favour of either arm in the absence of a true treatment effect. This serves as an important consistency check for the validity of the adaptive designs [8].

Accumulating evidence of a treatment difference motivates allocation to skew more towards the better arm. In the moderate effect scenario ($\theta_A = 0.65$), Simple RAR allocates 56.3% and Bayesian RAR 72.1% to arm A, compared to the fixed 50.0%. According to Equation 2, the theoretical Neyman optimal allocation at these parameter values is approximately 48.8%, meaning Simple RAR allocates slightly above this optimal ratio, whilst Bayesian RAR substantially exceeds it, driven by the Thompson Sampling mechanism concentrating allocation on the arm with the highest posterior probability of superiority [8]. Patient outcomes improve as a result: the mean number of successes per trial increases from 57.51 under Fixed 1:1 to 58.43 under Simple RAR and 60.70 under Bayesian RAR, representing approximately 3 additional successful outcomes per 100 participants under Bayesian RAR relative to the fixed design [4].

The ethical advantage is most pronounced under the large effect scenario ($\theta_A = 0.80$), as illustrated in Figure 2. Fixed 1:1 produces a single spike at exactly 50% with zero variability across replicates, directly reflecting its fixed allocation rule. Simple RAR spreads across approximately 55%–65%, centred at 58.8% with a standard deviation of 0.053, whilst Bayesian RAR concentrates the majority of its mass between 70% and 95%, with a mean allocation of 82.6% and a standard deviation of 0.070. The substantially broader spread of Bayesian RAR relative to Simple RAR is a direct manifestation of the continuously varying posterior-based allocation rule outlined in Section 4.3, whereby allocation responds proportionally to the accumulated evidence [8]. This yields a mean of 74.82 successes per trial under Bayesian RAR, compared to 67.52 under Simple RAR and 65.04 under Fixed 1:1, representing approximately 10 additional successful outcomes per 100 participants relative to the fixed design [4]. As established in Section 6.2, however, this ethical gain comes at the cost of a considerable reduction in statistical power, a tension that is examined further in Section 9 [4, 7].

6.4 Bayesian Posterior Evolution

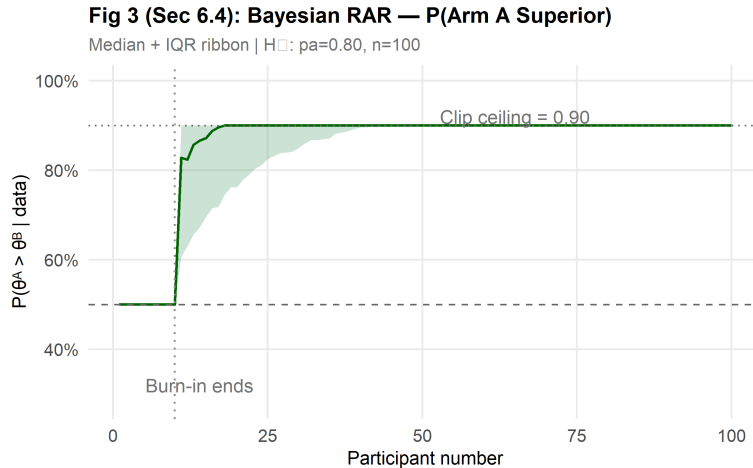


Figure 3: Median and interquartile range of the posterior probability $P(\theta_A > \theta_B \mid \text{data})$ across $B = 2000$ simulation replicates under the large effect scenario ($\theta_A = 0.80$, $\theta_B = 0.50$, $n = 100$). The dotted vertical line marks the end of the burn-in period at participant 10. The dotted horizontal line indicates the clip ceiling of 0.90.

Within the large effect scenario ($\theta_A = 0.80$), the posterior probability of arm A being superior, $P(\theta_A > \theta_B \mid \text{data})$, is shown in Figure 3 to evolve during the trial. For the burn-in period of the initial

10 participants, as per Section 3.4, the allocation probability is fixed at 0.50 regardless of the outcome. After burn-in, the median posterior probability rapidly increases to around 85% by participant 20 and hits the clip ceiling of 0.90 by around participant 25 [8]. The interquartile range ribbon is wide just after burn-in ends, indicating different speeds of learning across replicates, but this ribbon quickly narrows as the posterior consistently identifies arm A as the better treatment across all replicates [8].

The results presented in Sections 6.2 and 6.3 are explained directly by this pattern. Since the posterior reaches the clip ceiling so quickly, for the majority of the trial — roughly 75 out of 100 participants — the allocation probability is fixed at 0.90, which produces the high mean allocation of 82.6% to arm A reported in Section 6.3 [8]. This sustained imbalance means that arm B receives very little data for most of the trial, increasing $\text{Var}(\hat{\theta}_A - \hat{\theta}_B)$ as described in Equation 3, and causing the 15 percentage point reduction in power relative to Fixed 1:1 identified in Section 6.2 [4, 7].

6.5 Sensitivity Analysis Results

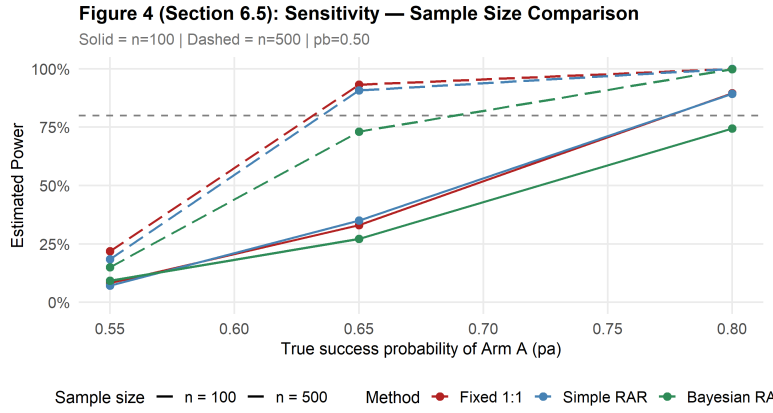


Figure 4: Power curves for Fixed 1:1, Simple RAR, and Bayesian RAR at $n = 100$ (solid) and $n = 500$ (dashed), with $\theta_B = 0.50$.

Table 1 and Figure 4 present the findings of the sensitivity analysis at $n = 500$. For the large treatment difference scenario ($\theta_A = 0.80$), the power deficit between Fixed 1:1 and Bayesian RAR almost completely vanishes — Fixed 1:1 and Simple RAR both achieve 1.000, whilst Bayesian RAR achieves 0.998 — suggesting that the power cost of adaptive randomisation is negligible at larger sample sizes when the true treatment difference is large [3, 7].

When the treatment difference is moderate ($\theta_A = 0.65$), however, a gap persists. Fixed 1:1 achieves 0.932 and Simple RAR 0.908, whilst Bayesian RAR achieves only 0.731, approximately 20 percentage points lower than Fixed 1:1. This is larger than the 6 percentage point difference observed at $n = 100$, and occurs because at $n = 500$ the posterior builds evidence faster, driving allocation even more strongly towards arm A (84.2% compared to 72.1% at $n = 100$) and amplifying the variance inflation of Equation 3 [4, 7].

In summary, increasing sample size improves power for all three designs, but does not fully resolve the power disadvantage of Bayesian RAR at moderate effect sizes. The choice between fixed and adaptive designs therefore remains a genuine trade-off that cannot be resolved simply by increasing n [3, 7].

7 Cluster Randomisation

7.1 Motivation and Design

In cluster randomised studies based on individually randomised study designs described in Sections 4.1 through 4.3, the randomisation of treatment arms occurs at the individual level. In addition, the statistical frameworks of Sections 3.1 and 3.2 are dependent upon this form of randomisation. However, in a number of real-world applications, individual randomisation is neither practically feasible nor scientifically acceptable. For example, when the intervention occurs at the group level (e.g., implementation of a hygiene protocol across all beds in an entire hospital ward; vaccination of an entire school population; a prescriptive guideline implemented across all doctors in a General Practice), it is impossible to randomly assign individuals to different treatment arms within the same group. Moreover, if the participants in close proximity can influence one another’s outcome, then they violate the Stable Unit Treatment Value Assumption that provides the basis for standard randomisation-based inference [2, 10]. Therefore, the proper method of conducting experiments in these types of situations involves the random assignment of entire clusters, which by definition consist of naturally occurring collections of participants, into treatment arms where all members of a particular cluster receive the same treatment [2, 10].

This section examines cluster randomised experimental design utilising $C = 20$ clusters per treatment arm, with $k = 10$ participants per cluster (for a total of 200 participants over both arms). The previously identified moderate treatment effect scenario ($\theta_A = 0.65$, $\theta_B = 0.50$, $\delta = 0.15$) from Section 5.1 will be used in order to provide a baseline for comparison purposes with the previous individually based simulations presented in Section 6.2. The intra-cluster correlation coefficient ρ will be evaluated at five levels: $\{0.00, 0.02, 0.05, 0.10, 0.20\}$ to reflect the range of intra-cluster correlations commonly found in clinical and public health trials [2, 10]. When $\rho = 0.00$ the Beta-Binomial cluster model of Section 4.4 degenerates to independent Bernoulli sampling and therefore recovers the previously individually randomised case and allows for an internal validation of the simulation [2].

The key quantity governing the statistical cost of clustering is the design effect $\text{DEFF} = 1 + (k - 1)\rho$ derived in Section 4.4 [2, 10]. With $k = 10$, this yields design effects of 1.00, 1.18, 1.45, 1.90, and 2.80 for the five ICC values respectively, meaning the effective sample size relative to an individually randomised trial with the same total number of participants is reduced by factors ranging from 15% at $\rho = 0.02$ to 64% at $\rho = 0.20$. Even at the commonly encountered value of $\rho = 0.05$, the effective sample size falls from 200 to approximately

$$n_{\text{eff}} = \frac{Ck}{\text{DEFF}} = \frac{200}{1.45} \approx 138, \quad (45)$$

a substantial reduction [2, 10]. The simulation study in Section 7.2 examines whether this formula accurately predicts the power loss observed in practice, and Section 7.3 investigates whether an adaptive cluster design can recover some of that lost power by directing more clusters towards the superior treatment arm [2, 10].

It is worth noting that the choice of $C = 20$ clusters per arm is intentionally modest, reflecting the practical constraints of many real-world cluster randomised trials in which recruiting a large number of clusters is logistically challenging [10]. As will be seen in Section 7.3, this has important consequences for the performance of adaptive designs at the cluster level, since the Thompson Sampling mechanism of Section 3.4 requires sufficient clusters in both arms to support reliable posterior updating. The interaction between cluster count, cluster size, and ICC therefore represents a three-way design trade-off that does not arise in individually randomised trials, and which must be carefully considered when planning any cluster randomised study [2, 10].

7.2 Statistical Framework

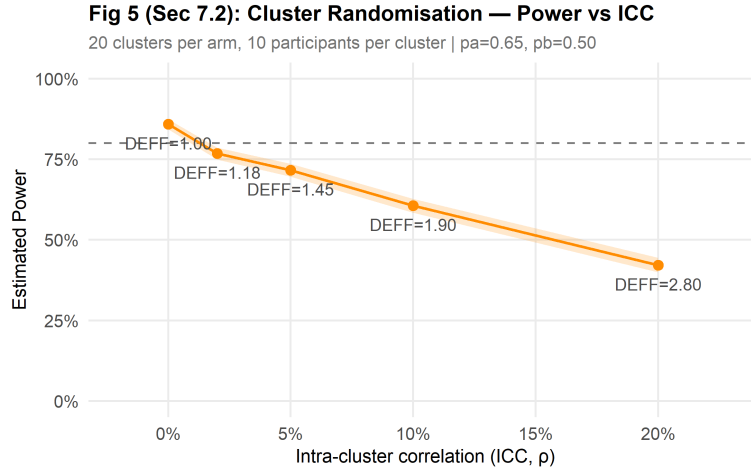


Figure 5: Estimated power of the fixed cluster randomised design as a function of ICC ρ , with $C = 20$ clusters per arm, $k = 10$ participants per cluster, $\theta_A = 0.65$, $\theta_B = 0.50$, $n = 200$, $B = 2000$ replicates. The dashed horizontal line indicates the conventional 80% power target. Design effect labels are shown at each ICC value.

Table 2 and Figure 5 present the estimated power of the fixed cluster randomised design across the five ICC values at $n = 200$ total participants. When $\rho = 0.00$, the design reduces to individual randomisation, yielding a power of 0.859 (95% CI: 0.843, 0.874), which comfortably exceeds the 80% target and confirms the internal consistency of the simulation [2]. As ρ increases, power decreases monotonically in close agreement with the theoretical predictions of Section 7.1 [2, 10]. At $\rho = 0.02$, power already falls below the 80% target to 0.768 (95% CI: 0.749, 0.786), and the degradation accelerates considerably at higher ICC values: at $\rho = 0.10$ power drops to 0.606 (95% CI: 0.584, 0.627), and at $\rho = 0.20$ it collapses to 0.422 (95% CI: 0.400, 0.444), meaning the trial retains less than half the power of the individually randomised baseline despite using the same total number of participants [2, 10].

Table 2: Estimated power of the fixed cluster randomised design across ICC values at $n = 200$ total participants ($C = 20$ clusters per arm, $k = 10$ participants per cluster), $\theta_A = 0.65$, $\theta_B = 0.50$, $B = 2000$ replicates. CI denotes 95% Monte Carlo confidence interval.

ICC (ρ)	Method	Power	MC SE	95% CI	DEFF	n_{eff}
0.00	Cluster Fixed	0.859	0.0078	(0.843, 0.874)	1.00	400.0
0.02	Cluster Fixed	0.768	0.0094	(0.749, 0.786)	1.18	339.0
0.05	Cluster Fixed	0.716	0.0101	(0.696, 0.736)	1.45	275.9
0.10	Cluster Fixed	0.606	0.0109	(0.584, 0.627)	1.90	210.5
0.20	Cluster Fixed	0.422	0.0110	(0.400, 0.444)	2.80	142.9

These results confirm that the design effect formula of Section 4.4 accurately captures the power loss attributable to clustering [2, 10]. The practical implication is significant: even a modest ICC of $\rho = 0.05$, which reduces the effective sample size from 400 to approximately 276, is sufficient to push power well below the 80% target [2, 10]. A trial designer who ignores the clustering structure and plans sample size as though participants were individually randomised will therefore systematically underpower the study, with the severity of this error growing rapidly as ρ increases [2, 10].

7.3 Simulation Extension

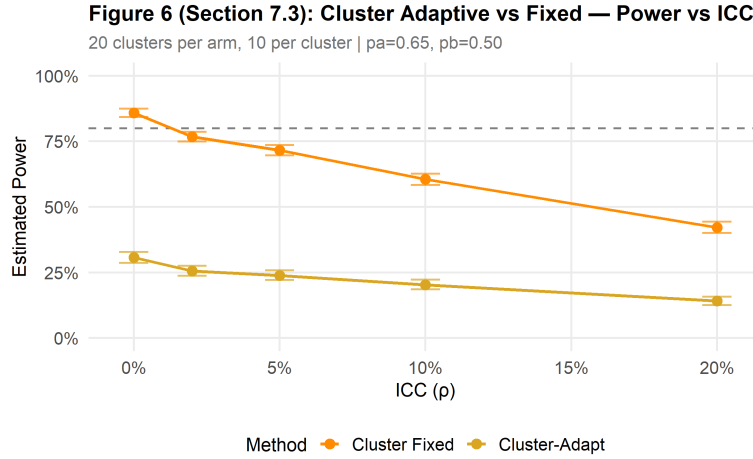


Figure 6: Estimated power of the fixed and adaptive cluster randomised designs as a function of ICC ρ , with $C = 20$ clusters per arm, $k = 10$ participants per cluster, $\theta_A = 0.65$, $\theta_B = 0.50$, $n = 200$, $B = 2000$ replicates. The dashed horizontal line indicates the conventional 80% power target.

Table 3 and Figure 6 compare the power of the fixed and adaptive cluster designs across the five ICC values. The results reveal a striking finding: contrary to the individual-level results of Section 6, where Bayesian RAR achieved meaningful ethical gains at a moderate power cost, the cluster adaptive design performs substantially worse than the fixed cluster design across every ICC value, and by a margin that is both large and statistically unambiguous [2, 8, 10].

At $\rho = 0.00$, the cluster adaptive design achieves a power of only 0.307 (95% CI: 0.287, 0.328), compared to 0.859 for the fixed cluster design, a reduction of over 55 percentage points whose confidence intervals do not overlap [2, 10]. This gap persists across all ICC values: at $\rho = 0.05$ the cluster adaptive design achieves 0.239 (95% CI: 0.221, 0.258) against 0.716 for the fixed design, and at $\rho = 0.20$ power falls to 0.141 (95% CI: 0.126, 0.157) against 0.422 for the fixed design [2, 10].

Table 3: Comparison of estimated power and allocation to arm A for the fixed and adaptive cluster randomised designs across ICC values at $n = 200$ total participants ($C = 20$ clusters per arm, $k = 10$ participants per cluster), $\theta_A = 0.65$, $\theta_B = 0.50$, $B = 2000$ replicates. CI denotes 95% Monte Carlo confidence interval.

ICC (ρ)	Method	Power	MC SE	95% CI	Alloc to A
0.00	Cluster Fixed	0.859	0.0078	(0.843, 0.874)	0.500
0.02	Cluster Fixed	0.768	0.0094	(0.749, 0.786)	0.500
0.05	Cluster Fixed	0.716	0.0101	(0.696, 0.736)	0.500
0.10	Cluster Fixed	0.606	0.0109	(0.584, 0.627)	0.500
0.20	Cluster Fixed	0.422	0.0110	(0.400, 0.444)	0.500
0.00	Cluster-Adapt	0.307	0.0103	(0.287, 0.328)	0.745
0.02	Cluster-Adapt	0.256	0.0098	(0.237, 0.275)	0.729
0.05	Cluster-Adapt	0.239	0.0095	(0.221, 0.258)	0.723
0.10	Cluster-Adapt	0.203	0.0090	(0.186, 0.222)	0.710
0.20	Cluster-Adapt	0.141	0.0078	(0.126, 0.157)	0.681

The cause of this severe power loss lies in the Thompson Sampling rule rapidly concentrating allocation towards the superior arm, with mean allocation to arm A ranging from 0.745 at $\rho = 0.00$ to 0.681 at $\rho = 0.20$ [8]. With only $C = 20$ clusters per arm, this skew leaves arm B with insufficient data to support reliable inference, inflating the variance of the estimated treatment difference as established in Equation 3 of Section 2.3 [7]. In such settings, fixed cluster randomisation with appropriate sample size adjustment for the expected ICC is therefore the preferable design choice [2, 10].

8 Covariate-Adjusted Adaptive Randomisation

8.1 Motivation

The individual-level adaptive designs of Sections 4.2 and 4.3 treat all participants as exchangeable, updating allocation probabilities based solely on the accumulated treatment outcomes without accounting for any participant characteristics that may influence the probability of a successful response. In practice, however, clinical trials frequently enrol participants who differ with respect to prognostic factors, such as age, disease severity, or comorbidity status, that are associated with the outcome independently of the treatment received [10]. Ignoring such covariates can reduce the precision of the estimated treatment effect and, in the context of adaptive randomisation, may lead to covariate imbalance between arms that distorts the posterior distributions driving the allocation rule [8].

Covariate-Adjusted Response-Adaptive Randomisation (CA-RAR) addresses this by incorporating covariate information directly into the allocation mechanism. Rather than updating posterior beliefs about θ_A and θ_B using raw outcome counts alone, CA-RAR fits a logistic regression model to the accumulated data at regular intervals, using the model to obtain covariate-adjusted estimates of each arm's success probability for the next participant. These adjusted estimates are then used to compute the posterior probability of superiority that drives the Thompson Sampling allocation rule of Section 3.4 [4, 8]. In this way, CA-RAR combines the ethical benefits of response-adaptive allocation with the inferential benefits of covariate adjustment, potentially improving both power and ethical allocation simultaneously [8].

The simulation study in this section examines whether CA-RAR offers any meaningful improvement over standard Bayesian RAR and Fixed 1:1 randomisation in terms of power, Type I error control, and ethical allocation [8]. A single binary covariate $X \sim \text{Bernoulli}(0.5)$ is included, with its effect on the outcome modelled on the log-odds scale via a coefficient β_x . The covariate effect is varied across $\{0.0, 0.3, 0.6, 1.0, 1.5\}$ to assess the conditions under which covariate adjustment confers a meaningful advantage. When $\beta_x = 0$ there is no covariate effect and CA-RAR reduces to standard Bayesian RAR, providing an internal consistency check for the simulation [8].

8.2 Covariate-Adjusted RAR

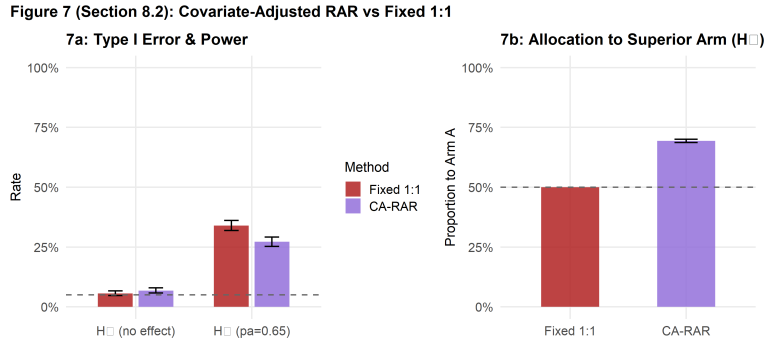


Figure 7: Type I error and power (panel 7a) and allocation to the superior arm under H_1 (panel 7b) for Fixed 1:1 and CA-RAR at $n = 100$, $\theta_A = 0.65$, $\theta_B = 0.50$, $\beta_x = 0.5$, $B = 2000$ replicates. The dashed horizontal lines indicate the nominal 5% significance level in panel 7a and 50% equal allocation in panel 7b.

Table 4 and Figure 7 present the estimated operating characteristics of CA-RAR and Fixed 1:1 randomisation under both the null and alternative hypotheses at $n = 100$ and $\beta_x = 0.5$. Under the null scenario ($\theta_A = \theta_B = 0.50$), both designs maintain adequate Type I error control. Fixed 1:1 achieves a rate of 0.056 (95% CI: 0.047, 0.067) and CA-RAR a rate of 0.068 (95% CI: 0.058, 0.080), with both confidence intervals including the nominal level of 0.05. The slightly elevated rate under CA-RAR is

consistent with the additional variability introduced by the adaptive allocation mechanism and the logistic regression refitting, but does not represent a systematic inflation of Type I error [8].

Under the alternative hypothesis ($\theta_A = 0.65$, $\theta_B = 0.50$), Fixed 1:1 achieves a power of 0.340 (95% CI: 0.319, 0.361) whilst CA-RAR achieves 0.272 (95% CI: 0.253, 0.292), a reduction of approximately 7 percentage points whose confidence intervals do not overlap. This power reduction mirrors the pattern observed for Bayesian RAR in Section 6.2 and arises for the same reason: CA-RAR allocates on average 69.4% of participants to arm A under H_1 , compared to the fixed 50.0%, increasing the variance of the estimated treatment difference as established in Equation 3 of Section 2.3 [4, 7]. The ethical benefit of this skew is reflected in the mean number of successes per trial: 65.96 under CA-RAR compared to 57.31 under Fixed 1:1, representing approximately 9 additional successful outcomes per 100 participants [4].

Table 4: Estimated operating characteristics of Fixed 1:1 and CA-RAR under the null and alternative hypotheses at $n = 100$, $\beta_x = 0.5$, $B = 2000$ replicates. CI denotes 95% Monte Carlo confidence interval. Alloc SD is not defined for Fixed 1:1 by construction.

Scenario	Method	Power/Type I	MC SE	95% CI	Alloc to A
H_0 ($p_A = 0.50$)	Fixed 1:1	0.056	0.0051	(0.047, 0.067)	0.500
H_0 ($p_A = 0.50$)	CA-RAR	0.068	0.0056	(0.058, 0.080)	0.494
H_1 ($p_A = 0.65$)	Fixed 1:1	0.340	0.0106	(0.319, 0.361)	0.500
H_1 ($p_A = 0.65$)	CA-RAR	0.272	0.0100	(0.253, 0.292)	0.694

These results indicate that CA-RAR replicates the fundamental statistical and ethical trade-off of standard Bayesian RAR at the individual level. The covariate adjustment does not resolve the power reduction associated with adaptive allocation, since the primary driver of power loss is the variance inflation from allocation skew rather than covariate imbalance [4, 8]. The conditions under which covariate adjustment genuinely improves performance are examined further in Section 8.3 through the sensitivity analysis on β_x [10].

8.3 Simulation Illustration

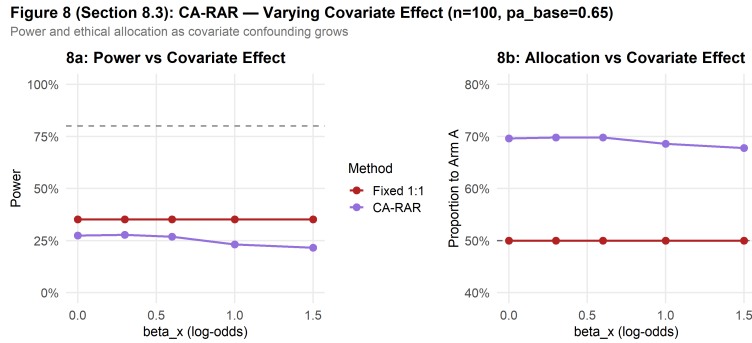


Figure 8: Estimated power (panel 8a) and allocation to arm A (panel 8b) for Fixed 1:1 and CA-RAR as the covariate effect β_x increases, at $n = 100$, $\theta_A = 0.65$, $\theta_B = 0.50$, $B = 2000$ replicates.

Table 5 and Figure 8 present the power and ethical allocation of CA-RAR and Fixed 1:1 as β_x increases from 0.0 to 1.5 on the log-odds scale. The power of Fixed 1:1 remains constant at 0.352 across all values of β_x , as expected since the fixed design does not use covariate information [7]. The power of CA-RAR is 0.274 at $\beta_x = 0.0$, consistent with the internal consistency check of Section 8.1, and remains broadly stable at $\beta_x = 0.3$ (0.278) and $\beta_x = 0.6$ (0.269) before declining to 0.232 at $\beta_x = 1.0$ and 0.216 at $\beta_x = 1.5$. Rather than improving with stronger covariate effects as might be anticipated, CA-RAR therefore loses power relative to Fixed 1:1 at all values of β_x , with the gap widening as the covariate effect grows [10].

Table 5: Estimated power and allocation to arm A for Fixed 1:1 and CA-RAR across covariate effect sizes β_x at $n = 100$, $\theta_A = 0.65$, $\theta_B = 0.50$, $B = 2000$ replicates.

β_x	Method	Power	Alloc to A
0.0	Fixed 1:1	0.352	0.500
0.0	CA-RAR	0.274	0.696
0.3	Fixed 1:1	0.352	0.500
0.3	CA-RAR	0.278	0.698
0.6	Fixed 1:1	0.352	0.500
0.6	CA-RAR	0.269	0.698
1.0	Fixed 1:1	0.352	0.500
1.0	CA-RAR	0.232	0.686
1.5	Fixed 1:1	0.352	0.500
1.5	CA-RAR	0.216	0.678

The explanation lies in the allocation mechanism. As β_x increases, the logistic regression model becomes increasingly confident in arm A’s superiority for covariate-adjusted participants, driving the allocation proportion to arm A from 0.696 at $\beta_x = 0.0$ to 0.678 at $\beta_x = 1.5$ [10]. This sustained skew amplifies the variance inflation of Equation 3, offsetting any precision gains from covariate adjustment [4, 7]. These findings suggest that CA-RAR, as implemented here, does not offer a statistical advantage over Fixed 1:1 at moderate sample sizes, and that the primary benefit of the covariate adjustment remains ethical rather than inferential [8, 10].

9 Discussion

9.1 Synthesis of Findings

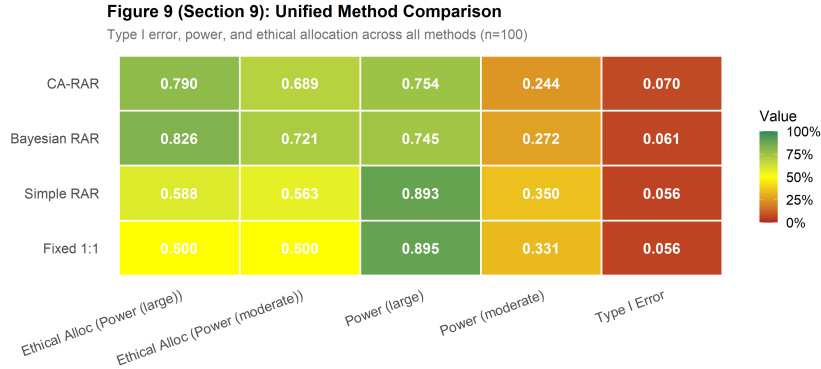


Figure 9: Unified comparison of Type I error, power, and ethical allocation across all four methods at $n = 100$. Green cells indicate higher values and red cells indicate lower values. Ethical allocation is reported as the mean proportion of participants allocated to the superior arm under H_1 .

Figure 9 and Table 6 provide a unified summary of the operating characteristics of all four methods across the null and alternative scenarios. The results reveal a clear and consistent pattern: no single method dominates across all three criteria simultaneously, and the choice of design necessarily involves a trade-off between statistical power and ethical allocation [4, 8].

Under the null hypothesis, all four methods maintain adequate Type I error control, with estimated rates of 0.056 for Fixed 1:1 and Simple RAR, 0.061 for Bayesian RAR, and 0.070 for CA-RAR. All confidence intervals include the nominal level of 0.05, confirming that none of the adaptive designs inflate Type I error in the absence of a true treatment effect [7, 8]. This is a non-trivial result: because adaptive allocation introduces dependence between successive assignments, maintaining valid

inference is not guaranteed without careful calibration of the burn-in period and clipping constraints of Section 3.4 [8].

Under the alternative hypothesis, the trade-off between power and ethical allocation becomes apparent. At the moderate effect size ($\theta_A = 0.65$), Fixed 1:1 achieves the highest power at 0.331, followed closely by Simple RAR at 0.350, whilst Bayesian RAR and CA-RAR achieve lower power of 0.272 and 0.244 respectively. However, Bayesian RAR and CA-RAR allocate substantially more participants to the superior arm, with mean proportions of 0.721 and 0.689 compared to the fixed 0.500 [4, 8]. At the large effect size ($\theta_A = 0.80$), Fixed 1:1 and Simple RAR achieve near-identical power of 0.895 and 0.893, whilst Bayesian RAR and CA-RAR achieve 0.745 and 0.754 respectively, with ethical allocations of 0.826 and 0.790. The sensitivity analysis of Section 6.5 demonstrated that this power gap narrows considerably at $n = 500$, suggesting that the cost of adaptive randomisation diminishes as sample size increases [7].

Table 6: Unified summary of Type I error, power, and ethical allocation across all four methods at $n = 100$, $B = 2000$ replicates. Ethical allocation is the mean proportion of participants allocated to the superior arm under H_1 . NA denotes not applicable.

Method	Scenario	Type I	Power	Ethical Alloc
Fixed 1:1	H_0	0.056	NA	0.500
Simple RAR	H_0	0.056	NA	0.499
Bayesian RAR	H_0	0.061	NA	0.493
CA-RAR	H_0	0.070	NA	0.499
Fixed 1:1	H_1 moderate ($p_A = 0.65$)	NA	0.331	0.500
Simple RAR	H_1 moderate ($p_A = 0.65$)	NA	0.350	0.563
Bayesian RAR	H_1 moderate ($p_A = 0.65$)	NA	0.272	0.721
CA-RAR	H_1 moderate ($p_A = 0.65$)	NA	0.244	0.689
Fixed 1:1	H_1 large ($p_A = 0.80$)	NA	0.895	0.500
Simple RAR	H_1 large ($p_A = 0.80$)	NA	0.893	0.588
Bayesian RAR	H_1 large ($p_A = 0.80$)	NA	0.745	0.826
CA-RAR	H_1 large ($p_A = 0.80$)	NA	0.754	0.790

Taken together, these findings confirm the central thesis of this report: fixed and adaptive randomisation represent genuinely different points on the statistical-ethical frontier, and the appropriate choice depends critically on the anticipated treatment difference and the relative weight placed on participant welfare versus inferential precision [4, 8]. Simple RAR occupies an interesting middle ground, achieving power essentially equivalent to Fixed 1:1 whilst providing modest ethical gains, making it an attractive compromise when the treatment difference is expected to be large [4]. Bayesian RAR and CA-RAR offer the strongest ethical benefits but at a meaningful power cost that must be carefully weighed against the clinical context of the trial [8].

9.2 Practical Implications for Trial Design

The findings of this report that has practical implications for the design of two arm clinical trials with binary outcome. The main point is that selection between fixed or adaptive randomisation should depend on size of treatment difference likely and whether participant welfare or power is prioritised [4, 7].

Using a fixed 1:1 randomisation is often a good idea when we anticipate the treatments to be similar, or when the trial is confirmatory. Because of these reasons, it optimises power at a fixed total sample size, randomisation-based inference is easy and regulators like the European Medicines Agency do not object to it (they view equal allocation as default for a confirmatory trial [7]). As shown in Section 6.1, the fixed 1:1 allocation maintains nominal Type I error control in all scenarios. Furthermore, the sensitivity analysis in Section 6.5 shows the gain in power of the fixed design over the adaptive designs is maximised at moderate effect sizes where the difference matters clinically the most [7].

Bayesian RAR will be a compelling alternative if a substantial treatment difference of clinical relevance is anticipated and the welfare of participants is a major consideration. Findings from Section 6.3 show that Bayesian RAR draws about 83% of participants to the successful arm under the large effect scenario. This is equivalent to an additional 10 successful outcomes per 100 trial participants relative to Fixed 1:1 [4]. The potential gain is large enough to be clinically meaningful. For example, if trial participants are suffering from a serious disease, allocating people to an inferior arm can involve a big human cost. The downside of this benefit must be offset against the power loss of approximately 15 percentage points which occurs at $n = 100$. Nonetheless, findings in Section 6.5 reveals that the disparity in terms of power lessens to a large extent at higher sample sizes [7].

A simple RAR offers a satisfactory compromise between power and ethics, presuming both are deemed important, but one must be moderately sacrificed. The way it allocates gives a maximum shift in favour of the better performing arm in probability of 0.60. This makes it have approximately the same power to Fixed 1:1 and the allocation is more ethically with modest gain over all effect sizes [4]. As a result, the strategy is mostly conservative and could prove most valuable in Phase II settings whose purpose is to identify a promising treatment for further examination rather than ultimate confirmation [7].

The cluster randomisation results in Section 7 give an important practical caveat: the benefits of adaptive allocation at the individual-level do not easily transfer to the cluster scenario. In particular, when there are few clusters, the Thompson Sampling procedure allocates too heavily to the better arm and does not leave enough data in the worse arm for reliable inference [2, 10]. Consequently, designers of adaptive cluster randomisation trials must examine whether the number of clusters available is sufficiently large for the allocation skew that the adaptive mechanism will induce. Moreover, before finalizing the design, power calculations through simulations should be performed under realistic ICC values [2, 10].

9.3 Limitations

There are some limitations of this research that should be taken into account. Firstly, this simulation study is limited to trials using binary outcomes, which is modelled by the Bernoulli distribution function. Therefore, it cannot be assumed that the conclusions drawn will apply equally well to trials using continuous outcomes, time-to-event data, or ordinal data, which have their own particular likelihood models and corresponding conjugate prior distributions that are different to the Beta-Binomial approach used in Section 3.3 [3, 6]. It is therefore reasonable to consider extending the simulations to cover the latter type of outcome.

Secondly, the sample sizes selected ($n = 100$ and $n = 500$) represent typical values for Phase II and Phase III trials respectively. However, we have not investigated how adaptive designs behave when sample size is very small, such as when working with rare diseases [3]. When sample sizes are extremely small, then the burn-in period of $b = 10$ participants that is incorporated into our algorithm will be a large fraction of the sample size overall; therefore, there could potentially be insufficient clipping to constrain the unstable allocations produced by the design during the early stages of a trial [8] (i.e., when $\epsilon = 0.1$).

Thirdly, the covariate adjusted design developed in Section 8 only included a single binary covariate, whereas real-world clinical trials usually include several prognostic variables of differing types [10]. Furthermore, our model was fitted to the data every five participants; although this was done for reasons of computational efficiency, it is possible that more frequent updating or use of Bayesian methods incorporating informative priors might produce somewhat different results. Thus, how CA-RAR performs on more complex covariate structures is another question still to be answered [10].

Therefore, the simulation assumes that outcomes can be observed immediately and that they can be used to determine each subsequent allocation. Unfortunately, this is rarely the case in practice. Allocation rules must often be determined before the start of a trial, since the response of a participant may not become known for many weeks or even months after treatment. Consequently, including delays in the results into the adaptive framework would require additional sophisticated methodologies, e.g. use of imputed interim responses, and thus also presents an important subject for further study [8].

10 Conclusion

This report looks at both the ethical and statistical trade-offs in choosing either adaptive or fixed randomisation when conducting two-arm clinical trials with binary endpoints using a Monte Carlo simulation framework written in R. In terms of a central conclusion, there was no single design that consistently performed well in all operating conditions simultaneously. Therefore, the correct design will be dependent on whether there is a small or large treatment effect and which of the ethical considerations (patient welfare vs. statistical power) are most important [4, 8].

In terms of achieving maximum power at a set sample size, fixed 1:1 randomisation is clearly the best option. As such it is the natural default for confirmatory trials, because regulatory requirements demand that equal numbers of patients are randomised to each treatment arm [7]. While RAR preserves nearly equivalent power compared to fixed 1:1 randomisation, it does provide some modest ethical advantages; therefore, it represents an appealing alternative in Phase II clinical trial settings [4]. Because it allocates approximately 83 percent of its participants to the better arm when the treatment effect is large, Bayesian RAR provides the greatest ethical advantage. It also generates approximately 10 more successful outcomes than fixed 1:1 randomization per 100 participants, with the loss in power being relatively minor as the sample size increases [4, 8].

While the cluster and covariate adjusted extensions demonstrated that the benefits of adaptive allocation do not translate directly into more complex trial designs, they were still able to show that there can be some potential benefits to adaptive allocation in these types of studies [2, 10]. However, with only $C = 20$ clusters per arm, the allocation of Thompson Sampling concentrated too heavily toward the superior arm, and thereby left too few data points for reliable inference [2, 10]. Similarly, while the covariate adjusted design provided some ethical advantages over standard Bayesian RAR at smaller sample sizes, the primary benefit of adjusting for covariates is likely going to be ethical, rather than inferential [10]. Future work may extend this type of framework to include continuous and time-to-event outcomes, multi-arm trials and delayed outcome settings, and build off of the simulation methods used here [8].

References

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A Appendix A: R Simulation Code

```
set.seed(123)
suppressPackageStartupMessages({
  library(ggplot2)
  library(dplyr)
  library(tidyr)
  library(patchwork)
  library(scales)
})

N_ITER <- 2000
N_POST <- 500

posterior_prob_A_superior <- function(successA, countA, successB, countB,
                                       alpha0 = 1, beta0 = 1,
                                       n_samples = N_POST) {
  sA <- rbeta(n_samples, alpha0 + successA, beta0 + (countA - successA))
  sB <- rbeta(n_samples, alpha0 + successB, beta0 + (countB - successB))
  mean(sA > sB)
}

mc_se <- function(p_hat, B) sqrt(p_hat * (1 - p_hat) / B)

run_prop_test <- function(assigned, observed, alternative = "two.sided") {
  tab <- table(factor(assigned, levels = c(0, 1)),
               factor(observed, levels = c(0, 1)))
  if (any(rowSums(tab) == 0)) return(1)
  prop.test(tab[, "1"], rowSums(tab),
            alternative = alternative,
            correct = FALSE)$p.value
}

.summarise_oc <- function(reject_null, alloc_a_prop, total_successes,
                         iterations, method = NULL) {
  n_rej <- sum(reject_null)
  p_hat <- mean(reject_null)
  alloc_mean <- mean(alloc_a_prop)
  alloc_sd <- sd(alloc_a_prop)
  alloc_ci <- alloc_mean + c(-1.96, 1.96) * alloc_sd / sqrt(iterations)

  if (!is.null(method) && identical(method, "Fixed 1:1")) {
    alloc_sd <- NA_real_
    alloc_ci <- c(NA_real_, NA_real_)
  }

  list(
    typeI_or_power = p_hat,
    mc_se = mc_se(p_hat, iterations),
    typeI_or_power_ci = prop.test(n_rej, iterations)$conf.int,
    alloc_mean = alloc_mean,
    alloc_sd = alloc_sd,
    alloc_ci = alloc_ci,
```

```

    successes_mean    = mean(total_successes),
    alloc_vals        = alloc_a_prop
  )
}

summarise_run <- function(res, scenario, method) {
  data.frame(
    Scenario          = scenario,
    Method             = method,
    Power_or_TypeI    = round(res$typeI_or_power, 3),
    MC_SE             = round(res$mc_se, 4),
    CI_L              = round(res$typeI_or_power_ci[1], 3),
    CI_U              = round(res$typeI_or_power_ci[2], 3),
    Alloc_to_A        = round(res$alloc_mean, 3),
    Alloc_SD          = ifelse(is.na(res$alloc_sd), NA, round(res$alloc_sd, 4)),
    Alloc_CI_L        = ifelse(is.na(res$alloc_ci[1]), NA, round(res$alloc_ci[1], 3)),
    Alloc_CI_U        = ifelse(is.na(res$alloc_ci[2]), NA, round(res$alloc_ci[2], 3)),
    Mean_Successes    = round(res$successes_mean, 2),
    stringsAsFactors = FALSE
  )
}

# 1) Fixed 1:1

run_fixed_sim <- function(n_total, pa, pb,
                          iterations = N_ITER,
                          alternative = "two.sided") {
  if (n_total %% 2 != 0) stop("n_total must be even for fixed 1:1.")
  reject_null    <- integer(iterations)
  alloc_a_prop    <- numeric(iterations)
  total_successes <- numeric(iterations)

  half <- n_total / 2L
  for (i in seq_len(iterations)) {
    assigned <- sample(c(rep(1L, half), rep(0L, half)))
    p_vec    <- ifelse(assigned == 1L, pa, pb)
    observed <- rbinom(n_total, 1L, p_vec)

    reject_null[i]    <- as.integer(
      run_prop_test(assigned, observed, alternative) < 0.05)
    alloc_a_prop[i]    <- mean(assigned)
    total_successes[i] <- sum(observed)
  }
  .summarise_oc(reject_null, alloc_a_prop, total_successes, iterations, "Fixed 1:1")
}

# 2) Simple RAR

run_rar_sim <- function(n_total, pa, pb,
                       iterations = N_ITER,
                       alternative = "two.sided",
                       p_up = 0.6, p_down = 0.4) {
  reject_null    <- integer(iterations)

```

```

alloc_a_prop    <- numeric(iterations)
total_successes <- numeric(iterations)

for (i in seq_len(iterations)) {
  assigned <- integer(n_total)
  observed <- integer(n_total)
  sA <- 0L; nA <- 0L; sB <- 0L; nB <- 0L

  for (j in seq_len(n_total)) {
    p_A <- if (nA == 0L || nB == 0L) 0.5 else {
      rA <- (sA + 0.5) / (nA + 1)
      rB <- (sB + 0.5) / (nB + 1)
      if (rA > rB) p_up else if (rB > rA) p_down else 0.5
    }
    assigned[j] <- rbinom(1L, 1L, p_A)
    observed[j] <- rbinom(1L, 1L, if (assigned[j] == 1L) pa else pb)
    if (assigned[j] == 1L) { nA <- nA + 1L; sA <- sA + observed[j]
    } else { nB <- nB + 1L; sB <- sB + observed[j] }
  }

  reject_null[i]    <- as.integer(
    run_prop_test(assigned, observed, alternative) < 0.05)
  alloc_a_prop[i]    <- mean(assigned)
  total_successes[i] <- sum(observed)
}

.summarise_oc(reject_null, alloc_a_prop, total_successes, iterations, "Simple RAR")
}

# 3) Bayesian RAR

run_brar_sim <- function(n_total, pa, pb,
                        iterations      = N_ITER,
                        alternative      = "two.sided",
                        alpha0 = 1, beta0 = 1,
                        burn_in         = 10,
                        n_posterior_samples = N_POST,
                        clip_lo = 0.1, clip_hi = 0.9,
                        store_trace = FALSE) {
  reject_null    <- integer(iterations)
  alloc_a_prop    <- numeric(iterations)
  total_successes <- numeric(iterations)
  posterior_trace <- if (store_trace) matrix(NA_real_, iterations, n_total) else NULL

  for (i in seq_len(iterations)) {
    assigned <- integer(n_total)
    observed <- integer(n_total)
    sA <- 0L; nA <- 0L; sB <- 0L; nB <- 0L

    for (j in seq_len(n_total)) {
      p_A <- if (j <= burn_in || nA == 0L || nB == 0L) 0.5 else {
        p_sup <- posterior_prob_A_superior(sA, nA, sB, nB,
                                           alpha0 = alpha0, beta0 = beta0,
                                           n_samples = n_posterior_samples)
      }
    }
  }
}

```

```

      pmax(clip_lo, pmin(clip_hi, p_sup))
    }
    if (store_trace) posterior_trace[i, j] <- p_A

    assigned[j] <- rbinom(1L, 1L, p_A)
    observed[j] <- rbinom(1L, 1L, if (assigned[j] == 1L) pa else pb)
    if (assigned[j] == 1L) { nA <- nA + 1L; sA <- sA + observed[j]
    } else { nB <- nB + 1L; sB <- sB + observed[j] }
  }

  reject_null[i] <- as.integer(
    run_prop_test(assigned, observed, alternative) < 0.05)
  alloc_a_prop[i] <- mean(assigned)
  total_successes[i] <- sum(observed)
}

oc <- .summarise_oc(reject_null, alloc_a_prop, total_successes,
  iterations, "Bayesian RAR")
oc$posterior_trace <- posterior_trace
oc
}

# 4) Cluster helpers

.beta_params_icc <- function(p, rho) {
  if (rho <= 0) return(list(a = Inf, b = Inf))
  phi <- (1 - rho) / rho
  list(a = p * phi, b = (1 - p) * phi)
}

run_cluster_sim <- function(m_clusters, k_per_clust, pa, pb,
  icc = 0.05, iterations = N_ITER) {
  bpA <- .beta_params_icc(pa, icc)
  bpB <- .beta_params_icc(pb, icc)
  reject_null <- integer(iterations)
  alloc_a_prop <- numeric(iterations)
  total_succ <- numeric(iterations)

  for (i in seq_len(iterations)) {
    muA <- if (icc <= 0) rep(pa, m_clusters) else rbeta(m_clusters, bpA$a, bpA$b)
    muB <- if (icc <= 0) rep(pb, m_clusters) else rbeta(m_clusters, bpB$a, bpB$b)
    propA <- vapply(muA, function(mu) mean(rbinom(k_per_clust, 1L, mu)), numeric(1))
    propB <- vapply(muB, function(mu) mean(rbinom(k_per_clust, 1L, mu)), numeric(1))

    total_succ[i] <- round(sum(c(propA, propB)) * k_per_clust)
    alloc_a_prop[i] <- 0.5
    p_val <- tryCatch(t.test(propA, propB, var.equal = FALSE)$p.value,
      error = function(e) NA_real_)
    reject_null[i] <- as.integer(!is.na(p_val) && p_val < 0.05)
  }

  deff <- 1 + (k_per_clust - 1) * icc
  oc <- .summarise_oc(reject_null, alloc_a_prop, total_succ,

```

```

        iterations, "Cluster Fixed")
oc$design_effect <- round(deff, 3)
oc$effective_n   <- round((m_clusters * k_per_clust * 2) / deff, 1)
oc
}

run_cluster_brar_sim <- function(m_clusters, k_per_clust, pa, pb,
                                icc = 0.05, iterations = N_ITER,
                                burn_in_clusters = 4,
                                n_posterior_samples = N_POST,
                                clip_lo = 0.1, clip_hi = 0.9) {
  bpA <- .beta_params_icc(pa, icc)
  bpB <- .beta_params_icc(pb, icc)
  reject_null <- integer(iterations)
  alloc_a_prop <- numeric(iterations)
  total_succ <- numeric(iterations)

  for (i in seq_len(iterations)) {
    arm_labels <- integer(m_clusters)
    clust_prop <- numeric(m_clusters)
    sA <- 0L; nA <- 0L; sB <- 0L; nB <- 0L

    for (cl in seq_len(m_clusters)) {
      p_A <- if (cl <= burn_in_clusters || nA == 0L || nB == 0L) 0.5 else {
        p_sup <- posterior_prob_A_superior(sA, nA, sB, nB,
                                           n_samples = n_posterior_samples)
        pmax(clip_lo, pmin(clip_hi, p_sup))
      }
      arm_labels[cl] <- rbinom(1L, 1L, p_A)
      true_p <- if (arm_labels[cl] == 1L) {
        if (icc <= 0) pa else rbeta(1L, bpA$a, bpA$b)
      } else {
        if (icc <= 0) pb else rbeta(1L, bpB$a, bpB$b)
      }
      y_cl <- rbinom(1L, size = k_per_clust, prob = true_p)
      clust_prop[cl] <- y_cl / k_per_clust

      if (arm_labels[cl] == 1L) { sA <- sA + y_cl; nA <- nA + k_per_clust
      } else { sB <- sB + y_cl; nB <- nB + k_per_clust }
    }

    propA <- clust_prop[arm_labels == 1L]
    propB <- clust_prop[arm_labels == 0L]
    alloc_a_prop[i] <- mean(arm_labels)
    total_succ[i] <- sum(round(clust_prop * k_per_clust))
    p_val <- tryCatch(
      if (length(propA) < 2 || length(propB) < 2) NA_real_
      else t.test(propA, propB, var.equal = FALSE)$p.value,
      error = function(e) NA_real_)
    reject_null[i] <- as.integer(!is.na(p_val) && p_val < 0.05)
  }
  .summarise_oc(reject_null, alloc_a_prop, total_succ, iterations, "Cluster-Adapt")
}

```

```
# 5) Covariate-adjusted RAR
```

```
run_ca_rar_sim <- function(n_total, pa_base, pb_base,
                           beta_x      = 0.5,
                           iterations   = N_ITER,
                           alternative   = "two.sided",
                           burn_in      = 20,
                           n_posterior_samples = N_POST,
                           clip_lo      = 0.1, clip_hi = 0.9,
                           refit_every  = 5) {
  logit      <- function(p) log(p / (1 - p))
  inv_logit  <- function(x) 1 / (1 + exp(-x))
  b0_A <- logit(pa_base)
  b0_B <- logit(pb_base)

  reject_null      <- integer(iterations)
  alloc_a_prop     <- numeric(iterations)
  total_successes  <- numeric(iterations)

  for (i in seq_len(iterations)) {
    assigned <- integer(n_total)
    observed <- integer(n_total)
    covar    <- rbinom(n_total, 1L, 0.5)
    sA <- 0L; nA <- 0L; sB <- 0L; nB <- 0L
    cached_p_A <- 0.5

    for (j in seq_len(n_total)) {
      xj <- covar[j]

      if (j <= burn_in || nA < 5L || nB < 5L) {
        p_A <- 0.5
      } else if ((j - burn_in) %% refit_every == 1L) {
        hist_df <- data.frame(Y = observed[seq_len(j - 1L)],
                              arm = assigned[seq_len(j - 1L)],
                              X = covar[seq_len(j - 1L)])
        fit <- tryCatch(glm(Y ~ arm + X, data = hist_df, family = binomial()),
                        error = function(e) NULL)
        if (is.null(fit)) {
          p_A <- cached_p_A
        } else {
          cf <- coef(fit)
          b_int <- unname(cf["(Intercept)"])
          b_arm <- unname(cf["arm"])
          b_x <- unname(cf["X"])
          if (anyNA(c(b_int, b_arm, b_x))) {
            p_A <- cached_p_A
          } else {
            eta_A <- b_int + b_arm * 1 + b_x * xj
            eta_B <- b_int + b_arm * 0 + b_x * xj
            p_adj_A <- inv_logit(eta_A)
            p_adj_B <- inv_logit(eta_B)
            eff_nA <- max(nA, 1L); eff_nB <- max(nB, 1L)
          }
        }
      }
    }
  }
}
```

```

    eff_sA <- round(p_adj_A * eff_nA)
    eff_sB <- round(p_adj_B * eff_nB)
    p_sup <- posterior_prob_A_superior(eff_sA, eff_nA, eff_sB, eff_nB,
                                       n_samples = n_posterior_samples)
    p_A <- pmax(clip_lo, pmin(clip_hi, p_sup))
    cached_p_A <- p_A
  }
}
} else {
  p_A <- cached_p_A
}

assigned[j] <- rbinom(1L, 1L, p_A)
true_p <- inv_logit(if (assigned[j] == 1L) b0_A + beta_x * xj
                    else b0_B + beta_x * xj)
observed[j] <- rbinom(1L, 1L, true_p)
if (assigned[j] == 1L) { nA <- nA + 1L; sA <- sA + observed[j]
} else { nB <- nB + 1L; sB <- sB + observed[j] }
}

alloc_a_prop[i] <- mean(assigned)
total_successes[i] <- sum(observed)
reject_null[i] <- as.integer(
  run_prop_test(assigned, observed, alternative) < 0.05)
}

.summarise_oc(reject_null, alloc_a_prop, total_successes, iterations, "CA-RAR")
}

# 6) Main suite

run_suite <- function(n_total = 100,
                     pb = 0.5,
                     pa_values = c(0.5, 0.55, 0.65, 0.8),
                     iterations = N_ITER) {
  out <- vector("list", length(pa_values) * 3L)
  k <- 0L
  for (pa in pa_values) {
    sc <- sprintf("pa=%.2f, pb=%.2f, n=%d", pa, pb, n_total)
    cat("Running:", sc, "\n")
    fixed <- run_fixed_sim(n_total, pa, pb, iterations)
    rar <- run_rar_sim(n_total, pa, pb, iterations)
    brar <- run_brar_sim(n_total, pa, pb, iterations, store_trace = FALSE)
    k <- k + 1L; out[[k]] <- summarise_run(fixed, sc, "Fixed 1:1")
    k <- k + 1L; out[[k]] <- summarise_run(rar, sc, "Simple RAR")
    k <- k + 1L; out[[k]] <- summarise_run(brar, sc, "Bayesian RAR")
  }
  do.call(rbind, out)
}

# Run simulations

cat("\n=== Section 6: Main suite n=100 ===\n")

```

```

results_100 <- run_suite(n_total = 100, pb = 0.5)

cat("\n=== Section 6.5: Sensitivity n=500 ===\n")
results_500 <- run_suite(n_total = 500, pb = 0.5)

cat("\n=== Section 7.2: Cluster fixed - varying ICC ===\n")
icc_vals <- c(0.00, 0.02, 0.05, 0.10, 0.20)
cluster_fixed_df <- do.call(rbind, lapply(icc_vals, function(rho) {
  cat(" ICC =", rho, "\n")
  res <- run_cluster_sim(20, 10, 0.65, 0.5, icc = rho)
  data.frame(ICC = rho, Method = "Cluster Fixed",
    Power          = round(res$typeI_or_power, 3),
    MC_SE          = round(res$mc_se, 4),
    CI_L           = round(res$typeI_or_power_ci[1], 3),
    CI_U           = round(res$typeI_or_power_ci[2], 3),
    Alloc_to_A     = round(res$alloc_mean, 3),
    Design_Effect  = round(res$design_effect, 3),
    Effective_N    = res$effective_n)
}))

cat("\n=== Section 7.3: Cluster adaptive vs fixed - varying ICC ===\n")
cluster_brar_df <- do.call(rbind, lapply(icc_vals, function(rho) {
  cat(" ICC =", rho, "\n")
  res <- run_cluster_brar_sim(20, 10, 0.65, 0.5, icc = rho)
  data.frame(ICC = rho, Method = "Cluster-Adapt",
    Power          = round(res$typeI_or_power, 3),
    MC_SE          = round(res$mc_se, 4),
    CI_L           = round(res$typeI_or_power_ci[1], 3),
    CI_U           = round(res$typeI_or_power_ci[2], 3),
    Alloc_to_A     = round(res$alloc_mean, 3),
    Design_Effect  = NA_real_,
    Effective_N    = NA_real_)
}))

cluster_all_df <- rbind(cluster_fixed_df, cluster_brar_df)

cat("\n=== Section 8.2: CA-RAR vs Fixed 1:1 ===\n")
ca_results <- rbind(
  summarise_run(run_fixed_sim(100, 0.5, 0.5), "H0 (pa=0.50)", "Fixed 1:1"),
  summarise_run(run_ca_rar_sim(100, 0.5, 0.5, beta_x = 0.5), "H0 (pa=0.50)", "CA-RAR"),
  summarise_run(run_fixed_sim(100, 0.65, 0.5), "H1 (pa=0.65)", "Fixed 1:1"),
  summarise_run(run_ca_rar_sim(100, 0.65, 0.5, beta_x = 0.5), "H1 (pa=0.65)", "CA-RAR")
)

cat("\n=== Section 8.3: CA-RAR - varying beta_x ===\n")
beta_x_vals <- c(0.0, 0.3, 0.6, 1.0, 1.5)
fixed_betax_res <- run_fixed_sim(100, 0.65, 0.5)
ca_betax_df <- do.call(rbind, lapply(beta_x_vals, function(bx) {
  cat(" beta_x =", bx, "\n")
  ca_res <- run_ca_rar_sim(100, 0.65, 0.5, beta_x = bx)
  rbind(
    data.frame(beta_x = bx, Method = "Fixed 1:1",
      Power = round(fixed_betax_res$typeI_or_power, 3),
      MC_SE = round(fixed_betax_res$mc_se, 4),

```

```

        Alloc = round(fixed_betax_res$alloc_mean, 3)),
data.frame(beta_x = bx, Method = "CA-RAR",
           Power = round(ca_res$typeI_or_power, 3),
           MC_SE = round(ca_res$mc_se, 4),
           Alloc = round(ca_res$alloc_mean, 3))
)
}))

cat("\n=== Section 9: Building discussion table ===\n")

.get_suite_res <- function(df, pa_str, method) {
  row <- df[grepl(pa_str, df$Scenario) & df$Method == method, ]
  list(typeI_or_power = row$Power_or_TypeI,
       alloc_mean      = row$Alloc_to_A)
}

cache_carar_H0   <- run_ca_rar_sim(100, 0.5, 0.5, beta_x = 0.5)
cache_carar_H165 <- run_ca_rar_sim(100, 0.65, 0.5, beta_x = 0.5)
cache_carar_H180 <- run_ca_rar_sim(100, 0.8, 0.5, beta_x = 0.5)

make_disc_row <- function(method, scenario, type1, power, alloc) {
  data.frame(Method = method, Scenario = scenario,
             Type_I = type1, Power = power,
             Ethical_Alloc = alloc, stringsAsFactors = FALSE)
}

.r <- function(x) round(x, 3)

discussion_table <- rbind(
  make_disc_row("Fixed 1:1", "H0",
               .r(.get_suite_res(results_100, "pa=0.50", "Fixed 1:1")$typeI_or_power),
               NA, .r(.get_suite_res(results_100, "pa=0.50", "Fixed 1:1")$alloc_mean)),
  make_disc_row("Simple RAR", "H0",
               .r(.get_suite_res(results_100, "pa=0.50", "Simple RAR")$typeI_or_power),
               NA, .r(.get_suite_res(results_100, "pa=0.50", "Simple RAR")$alloc_mean)),
  make_disc_row("Bayesian RAR", "H0",
               .r(.get_suite_res(results_100, "pa=0.50", "Bayesian RAR")$typeI_or_power),
               NA, .r(.get_suite_res(results_100, "pa=0.50",
                                     "Bayesian RAR")$alloc_mean)),
  make_disc_row("CA-RAR", "H0",
               .r(cache_carar_H0$typeI_or_power), NA, .r(cache_carar_H0$alloc_mean)),
  make_disc_row("Fixed 1:1", "H1 moderate (pa=0.65)", NA,
               .r(.get_suite_res(results_100, "pa=0.65", "Fixed 1:1")$typeI_or_power),
               .r(.get_suite_res(results_100, "pa=0.65", "Fixed 1:1")$alloc_mean)),
  make_disc_row("Simple RAR", "H1 moderate (pa=0.65)", NA,
               .r(.get_suite_res(results_100, "pa=0.65", "Simple RAR")$typeI_or_power),
               .r(.get_suite_res(results_100, "pa=0.65", "Simple RAR")$alloc_mean)),
  make_disc_row("Bayesian RAR", "H1 moderate (pa=0.65)", NA,
               .r(.get_suite_res(results_100, "pa=0.65", "Bayesian RAR")$typeI_or_power),
               .r(.get_suite_res(results_100, "pa=0.65", "Bayesian RAR")$alloc_mean)),
  make_disc_row("CA-RAR", "H1 moderate (pa=0.65)", NA,
               .r(cache_carar_H165$typeI_or_power), .r(cache_carar_H165$alloc_mean)),
  make_disc_row("Fixed 1:1", "H1 large (pa=0.80)", NA,

```

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        .r(.get_suite_res(results_100, "pa=0.80", "Fixed 1:1")$typeI_or_power),
        .r(.get_suite_res(results_100, "pa=0.80", "Fixed 1:1")$alloc_mean)),
make_disc_row("Simple RAR", "H1 large (pa=0.80)", NA,
        .r(.get_suite_res(results_100, "pa=0.80", "Simple RAR")$typeI_or_power),
        .r(.get_suite_res(results_100, "pa=0.80", "Simple RAR")$alloc_mean)),
make_disc_row("Bayesian RAR", "H1 large (pa=0.80)", NA,
        .r(.get_suite_res(results_100, "pa=0.80", "Bayesian RAR")$typeI_or_power),
        .r(.get_suite_res(results_100, "pa=0.80", "Bayesian RAR")$alloc_mean)),
make_disc_row("CA-RAR", "H1 large (pa=0.80)", NA,
        .r(cache_carar_H180$typeI_or_power), .r(cache_carar_H180$alloc_mean))
)

cat("\n=== Posterior trace for Figure 3 ===\n")
h1_brar_trace <- run_brar_sim(100, 0.8, 0.5, store_trace = TRUE)

df_trace <- data.frame(
  Participant = seq_len(100),
  Median = apply(h1_brar_trace$posterior_trace, 2, median, na.rm = TRUE),
  Q25 = apply(h1_brar_trace$posterior_trace, 2, quantile, 0.25, na.rm = TRUE),
  Q75 = apply(h1_brar_trace$posterior_trace, 2, quantile, 0.75, na.rm = TRUE)
)

cat("\n=== Allocation data for Figure 1 ===\n")
h1_fixed <- run_fixed_sim(100, 0.8, 0.5)
h1_rar <- run_rar_sim (100, 0.8, 0.5)
h1_brar <- run_brar_sim (100, 0.8, 0.5, store_trace = FALSE)

df_alloc <- data.frame(
  Method = factor(rep(c("Fixed 1:1", "Simple RAR", "Bayesian RAR"), each = N_ITER),
    levels = c("Fixed 1:1", "Simple RAR", "Bayesian RAR")),
  Proportion = c(h1_fixed$alloc_vals, h1_rar$alloc_vals, h1_brar$alloc_vals)
)

cat("\n--- Table 1: n=100 ---\n")
print(results_100, row.names = FALSE)
cat("\n--- Table 2: n=500 Sensitivity ---\n")
print(results_500, row.names = FALSE)
cat("\n--- Table 3: Cluster Fixed ---\n")
print(cluster_fixed_df, row.names = FALSE)
cat("\n--- Table 4: Cluster Adaptive vs Fixed ---\n")
print(cluster_all_df, row.names = FALSE)
cat("\n--- Table 5: CA-RAR vs Fixed ---\n")
print(ca_results, row.names = FALSE)
cat("\n--- Table 6: CA-RAR varying beta_x ---\n")
print(ca_betax_df, row.names = FALSE)
cat("\n--- Table 7: Discussion Table ---\n")
print(discussion_table, row.names = FALSE)

COLS <- c(
  "Fixed 1:1" = "firebrick",
  "Simple RAR" = "steelblue",
  "Bayesian RAR" = "seagreen",
  "CA-RAR" = "mediumpurple",

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  "Cluster Fixed" = "darkorange",
  "Cluster-Adapt" = "goldenrod"
)

BASE_THEME <- theme_minimal(base_size = 12) +
  theme(
    plot.title      = element_text(face = "bold", size = 13),
    plot.subtitle   = element_text(colour = "grey45", size = 10),
    legend.position = "bottom",
    panel.grid.minor = element_blank()
  )

# Figure 1
power_df <- results_100 %>%
  filter(!grepl("pa=0.50", Scenario)) %>%
  mutate(pa      = as.numeric(sub("pa=(.*?),.*", "\\1", Scenario)),
         Method = factor(Method, levels = c("Fixed 1:1", "Simple RAR", "Bayesian RAR")))

fig1 <- ggplot(power_df, aes(pa, Power_or_TypeI, colour = Method, group = Method)) +
  geom_hline(yintercept = 0.8, linetype = "dashed", colour = "grey50") +
  geom_line(linewidth = 1.1) + geom_point(size = 3) +
  geom_errorbar(aes(ymin = CI_L, ymax = CI_U), width = 0.012, alpha = 0.6) +
  scale_colour_manual(values = COLS) +
  scale_y_continuous(labels = percent_format(1), limits = c(0, 1)) +
  BASE_THEME +
  labs(title      = "Figure 1 (Section 6.2): Power Curves by Method (n=100, pb=0.50)",
       subtitle   = "Error bars = 95% Monte Carlo CI | dashed = 80% conventional target",
       x = "True success probability of Arm A (pa)",
       y = "Estimated Power", colour = "Method")
ggsave("fig1_power_curves.png", fig1, width = 8, height = 5, dpi = 300)

# Figure 2
fig2 <- ggplot(df_alloc, aes(x = Proportion, fill = Method)) +
  geom_histogram(alpha = 0.55, position = "identity", bins = 35) +
  geom_vline(xintercept = 0.5, linetype = "dashed", colour = "grey30") +
  scale_fill_manual(values = COLS) +
  scale_x_continuous(labels = percent_format(1)) +
  BASE_THEME +
  labs(title      = "Figure 2 (Section 6.3): Allocation to Arm A Under H\u2081",
       subtitle   = "pa=0.80, pb=0.50, n=100 | dashed = 50% equal allocation",
       x = "Proportion allocated to Arm A", y = "Frequency", fill = "Method")
ggsave("fig2_allocation.png", fig2, width = 8, height = 5, dpi = 300)

# Figure 3
fig3 <- ggplot(df_trace, aes(x = Participant)) +
  geom_ribbon(aes(ymin = Q25, ymax = Q75), fill = "seagreen", alpha = 0.25) +
  geom_line(aes(y = Median), colour = "darkgreen", linewidth = 1) +
  geom_hline(yintercept = c(0.5, 0.9),
            linetype = c("dashed", "dotted"), colour = "grey40", linewidth = 0.6) +
  geom_vline(xintercept = 10, linetype = "dotted", colour = "grey55") +
  annotate("text", x = 14, y = 0.32, label = "Burn-in ends",
         colour = "grey45", size = 3) +
  annotate("text", x = 65, y = 0.92, label = "Clip ceiling = 0.90",

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    colour = "grey45", size = 3) +
  scale_y_continuous(limits = c(0.28, 1), labels = percent_format(1)) +
  BASE_THEME +
  labs(title = paste0("Figure 3 (Section 6.4): Bayesian RAR",
    " \u2014 P(Arm A Superior) Over Trial"),
    subtitle = "Median + IQR ribbon | H\u2081: pa=0.80, n=100",
    x = "Participant number", y = "P(\u03b8\u1d2c > \u03b8\u1d2e | data)")
ggsave("fig3_posterior.png", fig3, width = 8, height = 5, dpi = 300)

# Figure 4
.prep_sens <- function(df, n_label) {
  df %>%
    filter(!grepl("pa=0.50", Scenario)) %>%
    mutate(pa = as.numeric(sub("pa=(.*?),.*", "\\1", Scenario)),
      n_label = n_label,
      Method = factor(Method, c("Fixed 1:1", "Simple RAR", "Bayesian RAR")))
}
sens_df <- bind_rows(.prep_sens(results_100, "n = 100"),
  .prep_sens(results_500, "n = 500"))

fig4 <- ggplot(sens_df, aes(pa, Power_or_TypeI,
  colour = Method,
  linetype = n_label,
  group = interaction(Method, n_label))) +
  geom_hline(yintercept = 0.8, linetype = "dashed", colour = "grey50") +
  geom_line(linewidth = 1) + geom_point(size = 2.5) +
  scale_colour_manual(values = COLS) +
  scale_linetype_manual(values = c("n = 100" = "solid", "n = 500" = "longdash")) +
  scale_y_continuous(labels = percent_format(1), limits = c(0, 1)) +
  BASE_THEME +
  labs(title = "Figure 4 (Section 6.5): Sensitivity \u2014 Sample Size Comparison",
    subtitle = "Solid = n=100 | Dashed = n=500 | pb=0.50",
    x = "True success probability of Arm A (pa)", y = "Estimated Power",
    colour = "Method", linetype = "Sample size")
ggsave("fig4_sensitivity.png", fig4, width = 9, height = 5, dpi = 300)

# Figure 5
fig5 <- ggplot(cluster_fixed_df, aes(ICC, Power)) +
  geom_hline(yintercept = 0.8, linetype = "dashed", colour = "grey50") +
  geom_ribbon(aes(ymin = CI_L, ymax = CI_U), fill = "darkorange", alpha = 0.2) +
  geom_line(colour = "darkorange", linewidth = 1.1) +
  geom_point(colour = "darkorange", size = 3) +
  geom_text(aes(label = sprintf("DEFF=%.2f", Design_Effect)),
    vjust = -1, size = 3, colour = "grey30") +
  scale_x_continuous(labels = percent_format(1)) +
  scale_y_continuous(labels = percent_format(1), limits = c(0, 1)) +
  BASE_THEME +
  labs(title = "Figure 5 (Section 7.2): Cluster Randomisation \u2014 Power vs ICC",
    subtitle = "20 clusters per arm, 10 participants per cluster | pa=0.65, pb=0.50",
    x = "Intra-cluster correlation (ICC, \u03c1)", y = "Estimated Power")
ggsave("fig5_cluster_power.png", fig5, width = 8, height = 5, dpi = 300)

```

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# Figure 6
fig6 <- cluster_all_df %>%
  mutate(Method = factor(Method, levels = c("Cluster Fixed", "Cluster-Adapt"))) %>%
  ggplot(aes(ICC, Power, colour = Method, group = Method)) +
  geom_hline(yintercept = 0.8, linetype = "dashed", colour = "grey50") +
  geom_line(linewidth = 1.1) + geom_point(size = 3) +
  geom_errorbar(aes(ymin = CI_L, ymax = CI_U), width = 0.01, alpha = 0.7) +
  scale_colour_manual(values = c("Cluster Fixed" = "darkorange",
                                "Cluster-Adapt" = "goldenrod")) +
  scale_x_continuous(labels = percent_format(1)) +
  scale_y_continuous(labels = percent_format(1), limits = c(0, 1)) +
  BASE_THEME +
  labs(title = "Figure 6 (Section 7.3): Cluster Adaptive vs Fixed \u2014 Power vs ICC",
       subtitle = "20 clusters per arm, 10 per cluster | pa=0.65, pb=0.50",
       x = "ICC (\u03c1)", y = "Estimated Power", colour = "Method")
ggsave("fig6_cluster_adapt.png", fig6, width = 8, height = 5, dpi = 300)

# Figure 7
ca_long <- ca_results %>%
  mutate(Method = factor(Method, c("Fixed 1:1", "CA-RAR")),
         Hypothesis = ifelse(grepl("H0", Scenario),
                              "H\u2080 (no effect)", "H\u2081 (pa=0.65)"))

f7a <- ggplot(ca_long, aes(Hypothesis, Power_or_TypeI, fill = Method)) +
  geom_col(position = position_dodge(0.7), width = 0.6, alpha = 0.85) +
  geom_errorbar(aes(ymin = CI_L, ymax = CI_U),
               position = position_dodge(0.7), width = 0.25, linewidth = 0.7) +
  geom_hline(yintercept = 0.05, linetype = "dashed", colour = "grey40") +
  scale_fill_manual(values = c("Fixed 1:1" = "firebrick", "CA-RAR" = "mediumpurple")) +
  scale_y_continuous(labels = percent_format(1), limits = c(0, 1)) +
  BASE_THEME + theme(legend.position = "right") +
  labs(title = "7a: Type I Error & Power", x = NULL, y = "Rate", fill = "Method")

f7b <- ca_long %>% filter(grepl("H1", Scenario)) %>%
  ggplot(aes(Method, Alloc_to_A, fill = Method)) +
  geom_col(width = 0.5, alpha = 0.85) +
  geom_errorbar(aes(ymin = Alloc_CI_L, ymax = Alloc_CI_U), width = 0.2, linewidth = 0.7) +
  geom_hline(yintercept = 0.5, linetype = "dashed", colour = "grey40") +
  scale_fill_manual(values = c("Fixed 1:1" = "firebrick", "CA-RAR" = "mediumpurple")) +
  scale_y_continuous(labels = percent_format(1), limits = c(0, 1)) +
  BASE_THEME + theme(legend.position = "none") +
  labs(title = "7b: Allocation to Superior Arm (H\u2081)",
       x = NULL, y = "Proportion to Arm A")

fig7 <- f7a + f7b + plot_annotation(
  title = "Figure 7 (Section 8.2): Covariate-Adjusted RAR vs Fixed 1:1",
  theme = theme(plot.title = element_text(face = "bold", size = 13))
)
ggsave("fig7_ca_rar.png", fig7, width = 11, height = 5, dpi = 300)

```

```

# Figure 8
ca_betax_plot <- ca_betax_df %>%
  mutate(Method = factor(Method, c("Fixed 1:1", "CA-RAR")))

f8a <- ggplot(ca_betax_plot, aes(beta_x, Power, colour = Method, group = Method)) +
  geom_hline(yintercept = 0.8, linetype = "dashed", colour = "grey50") +
  geom_line(linewidth = 1.1) + geom_point(size = 3) +
  scale_colour_manual(values = c("Fixed 1:1" = "firebrick", "CA-RAR" = "mediumpurple")) +
  scale_y_continuous(labels = percent_format(1), limits = c(0, 1)) +
  BASE_THEME + theme(legend.position = "right") +
  labs(title = "8a: Power vs Covariate Effect",
       x = "beta_x (log-odds)", y = "Power", colour = "Method")

f8b <- ggplot(ca_betax_plot, aes(beta_x, Alloc, colour = Method, group = Method)) +
  geom_hline(yintercept = 0.5, linetype = "dashed", colour = "grey40") +
  geom_line(linewidth = 1.1) + geom_point(size = 3) +
  scale_colour_manual(values = c("Fixed 1:1" = "firebrick", "CA-RAR" = "mediumpurple")) +
  scale_y_continuous(labels = percent_format(1), limits = c(0.4, 0.8)) +
  BASE_THEME + theme(legend.position = "none") +
  labs(title = "8b: Allocation to Arm A vs Covariate Effect",
       x = "beta_x (log-odds)", y = "Proportion to Arm A")

fig8 <- f8a + f8b + plot_annotation(
  title = paste0("Figure 8 (Section 8.3): CA-RAR \u2014 Varying",
                 " Covariate Effect (n=100, pa_base=0.65)"),
  subtitle = "Power and ethical allocation as covariate confounding grows",
  theme = theme(plot.title = element_text(face = "bold", size = 13),
                plot.subtitle = element_text(colour = "grey45", size = 10))
)
ggsave("fig8_ca_betax.png", fig8, width = 11, height = 5, dpi = 300)

# Figure 9
disc_heat <- discussion_table %>%
  mutate(
    Metric = case_when(
      Scenario == "H0" ~ "Type I Error",
      Scenario == "H1 moderate (pa=0.65)" ~ "Power (moderate)",
      Scenario == "H1 large (pa=0.80)" ~ "Power (large)"
    ),
    Value = ifelse(!is.na(Power), Power, Type_I),
    Method = factor(Method, levels = c("Fixed 1:1", "Simple RAR",
                                       "Bayesian RAR", "CA-RAR"))
  ) %>%
  filter(!is.na(Value))

disc_long <- bind_rows(
  disc_heat %>% transmute(Method, Metric, Value),
  disc_heat %>%
    filter(Metric != "Type I Error") %>%
    transmute(Method,
              Metric = paste0("Ethical Alloc (", Metric, ")"),
              Value = Ethical_Alloc)
)

```

```

fig9 <- ggplot(disc_long, aes(x = Metric, y = Method, fill = Value)) +
  geom_tile(colour = "white", linewidth = 0.8) +
  geom_text(aes(label = sprintf("%.3f", Value)),
    size = 3.5, colour = "white", fontface = "bold") +
  scale_fill_gradient2(low = "firebrick", mid = "yellow", high = "seagreen",
    midpoint = 0.5, limits = c(0, 1),
    labels = percent_format(1)) +
  scale_x_discrete(guide = guide_axis(angle = 20)) +
  BASE_THEME +
  theme(legend.position = "right", panel.grid = element_blank()) +
  labs(title = "Figure 9 (Section 9): Unified Method Comparison",
    subtitle = paste0("Type I error, power, and ethical allocation",
      " across all methods (n=100)"),
    x = NULL, y = NULL, fill = "Value")
ggsave("fig9_discussion_heatmap.png", fig9, width = 11, height = 5, dpi = 300)

cat("\n=== All 9 figures and 7 tables complete. ===\n")

```